NORMS IN LINGUISTIC HUMAN-AI INTERACTION

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GOALS

CHANSE.
COLLABORATION OF HUMANITIES AND SOCIAL SCIENCES IN EUROPE

HERA
Humanities in the European Research Area
CHANSE-CR-939 NIHAI

1. Investigate norms of linguistic communication among humans and in and human-AI interaction.
2. Devise principles for the responsible design for trustworthy chatbots.

www.talkingtobots.net

NORMS OF ASSERTION

Assertions: Speech acts by means of which we share information.

Q: In what epistemic condition must a speaker be to assert *p*?

KEY QUESTION

- When can you make a certain claim *p*? (e.g. 'It's raining in Paris.')

ACCOUNTS

- *Belief*: Assert that *p* only if you believe that *p*.
  - Bach 2008, Bach & Harnish 1979

ACCOUNTS

- *Belief*
- *Justification*: Assert p only if you are justified in asserting p.
  - Douven 2006, Lackey 2007

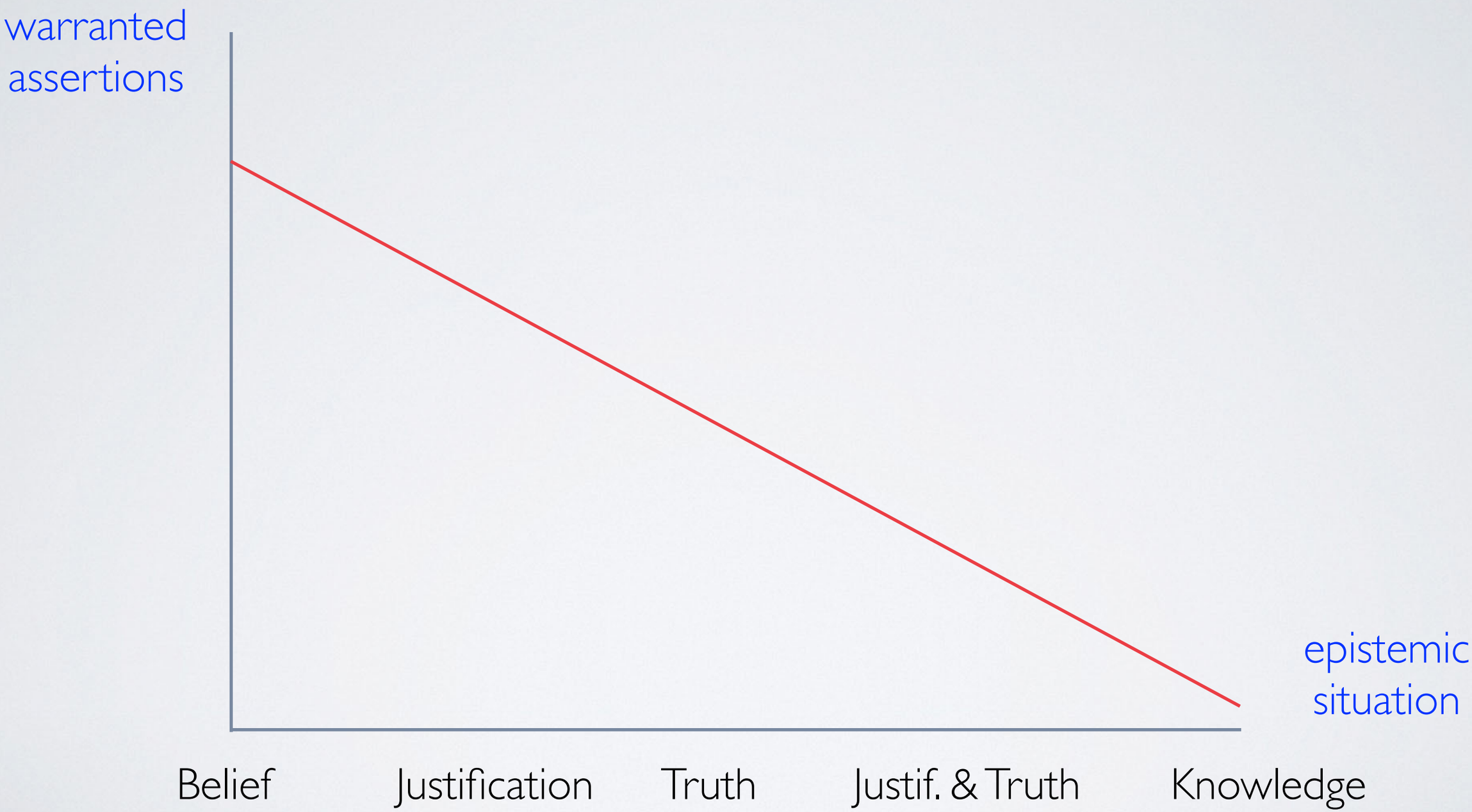
ACCOUNTS

- *Belief*
- *Justified Belief*
- *Truth*: Assert that *p* only if *p* is true.
  - Weiner 2005; cf. also Dummett 1959

ACCOUNTS

- *Belief*
- *Justified Belief*
- *Truth*
- *Knowledge*: Assert that *p* only if you know that *p*.
  - Williamson, 1996, 2002, Brandom 1998, DeRose 1996, Adler 2002, Hawthorne 2003, Garcia-Carpintero, 2004, Turri 2011, Benton 2011

NORM OF ASSERTION

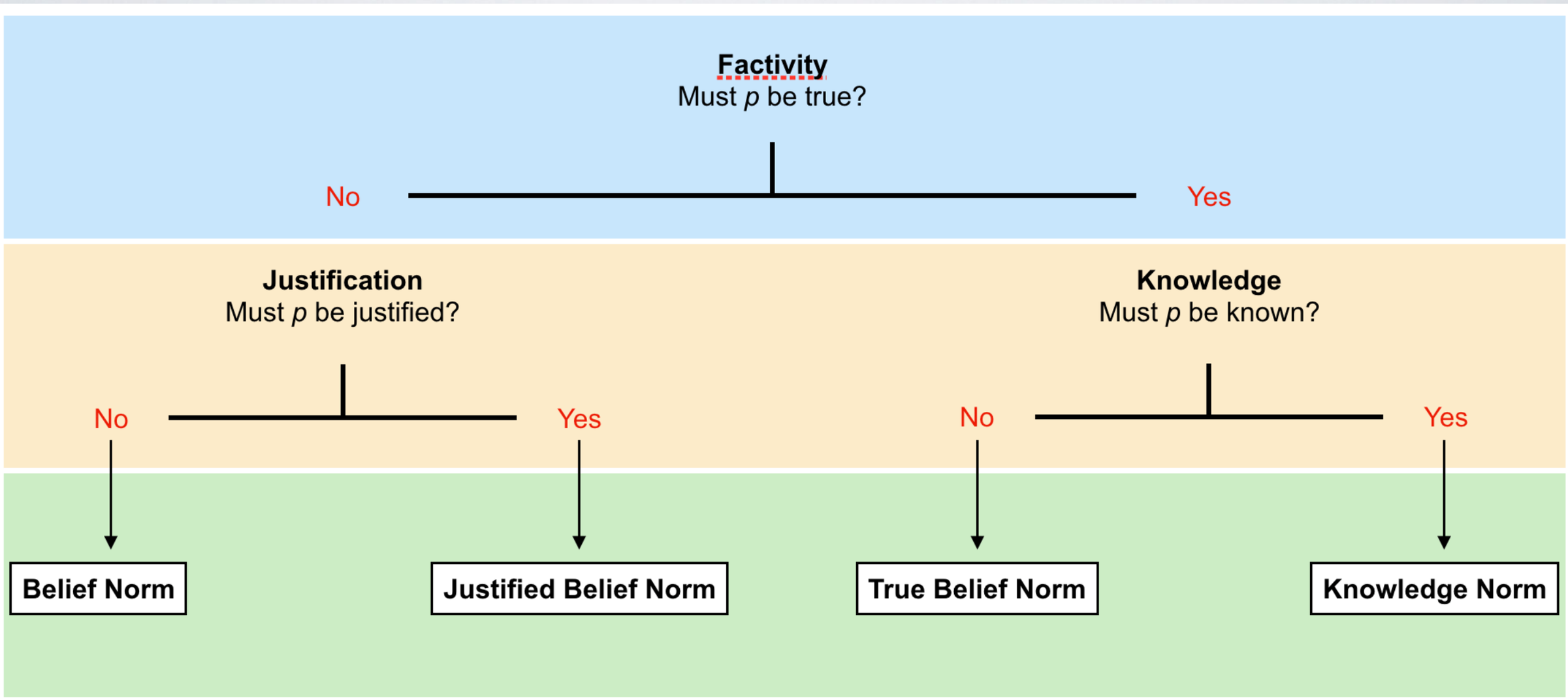


AN EMPIRICAL QUESTION

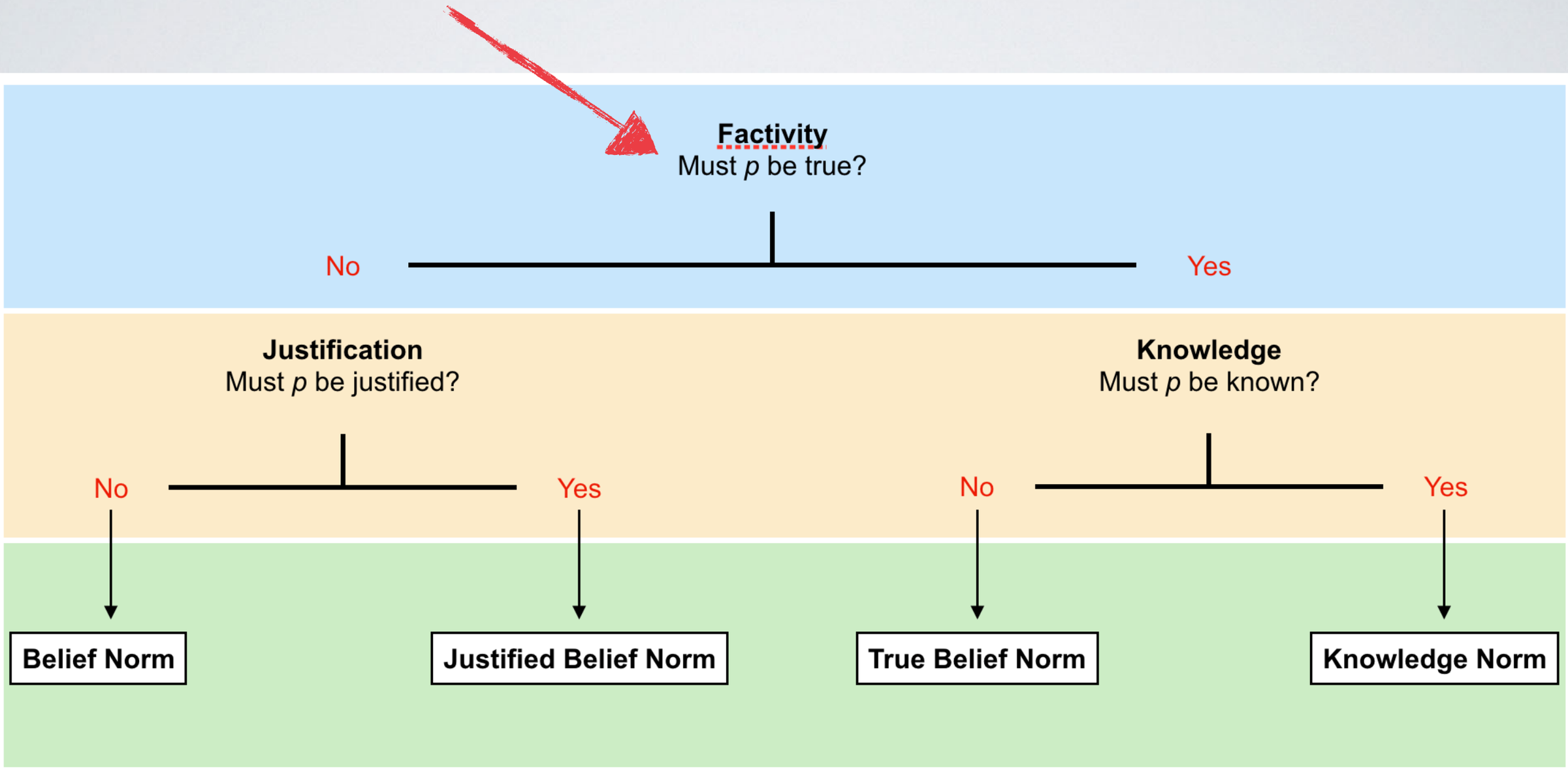
- The question concerning the norm of assertion is an empirical question (Pagin, 2016; Turri, 2014).
- There is only so much we can do from the armchair, in particular since these norms are not explicitly defined.

NORMS OF HUMAN ASSERTION

GAME PLAN


Factivity
Must p be true?
No
Yes
Justification
Must p be justified?
No
Yes
Knowledge
Must p be known?
No
Yes
Belief Norm
Justified Belief Norm
True Belief Norm
Knowledge Norm

GAME PLAN


Factivity
Must p be true?
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Justified Belief Norm
True Belief Norm
Knowledge Norm

SCENARIO

TRUE	FALSE
Bob has a friend, Jill, who has driven a Buick for many years. A Buick is an American car. Bob therefore thinks that Jill drives an American car.	
Jill still drives a Buick.	Unbeknownst to Bob, Jill now drives a Mercedes.

SCENARIO

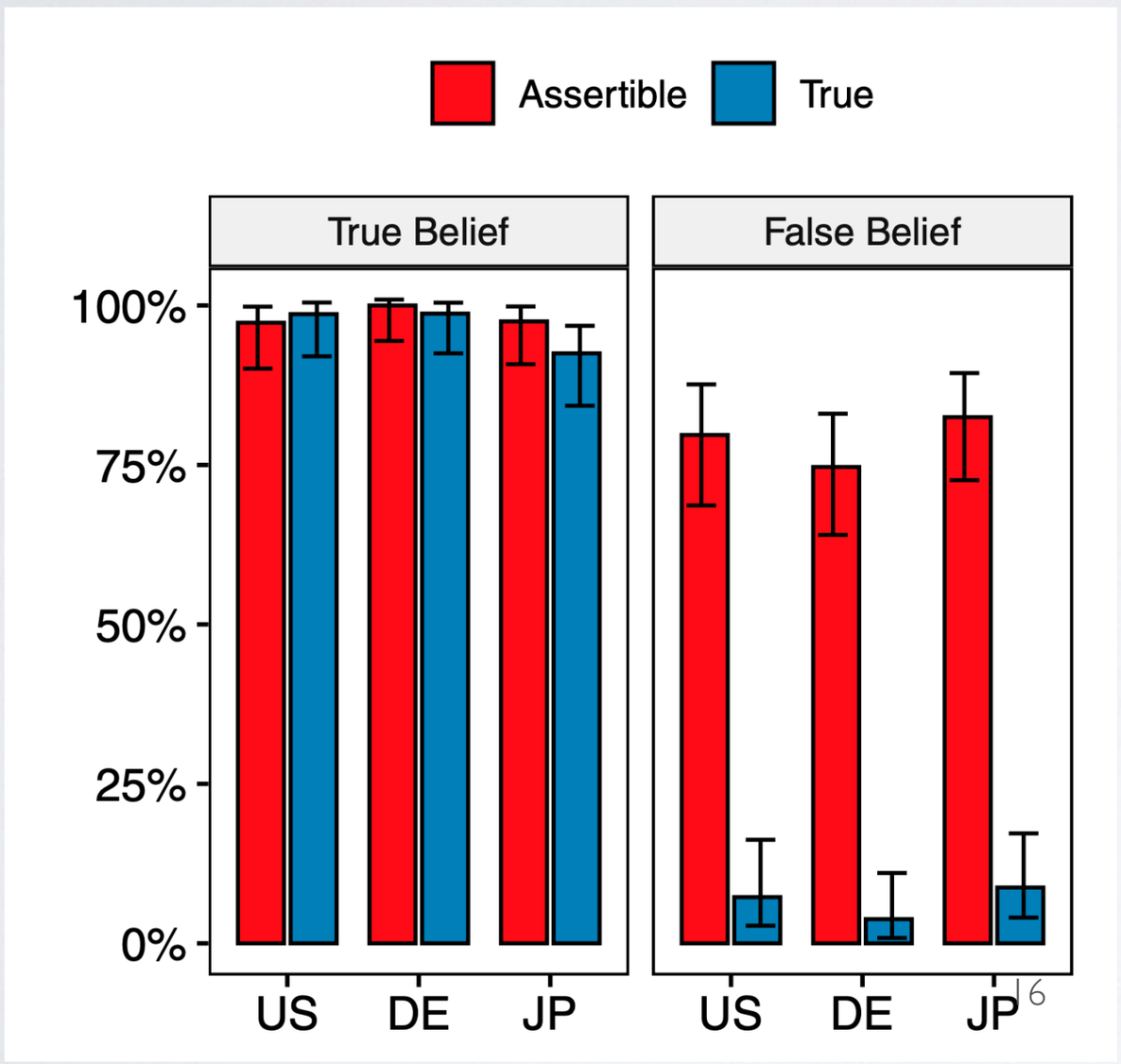
Q1: Should Bob say Jill owns an American car? (Yes/No)

Q2: Is it true that Jill owns an American car? (Yes/No)

FACTIVITY

461 (495)

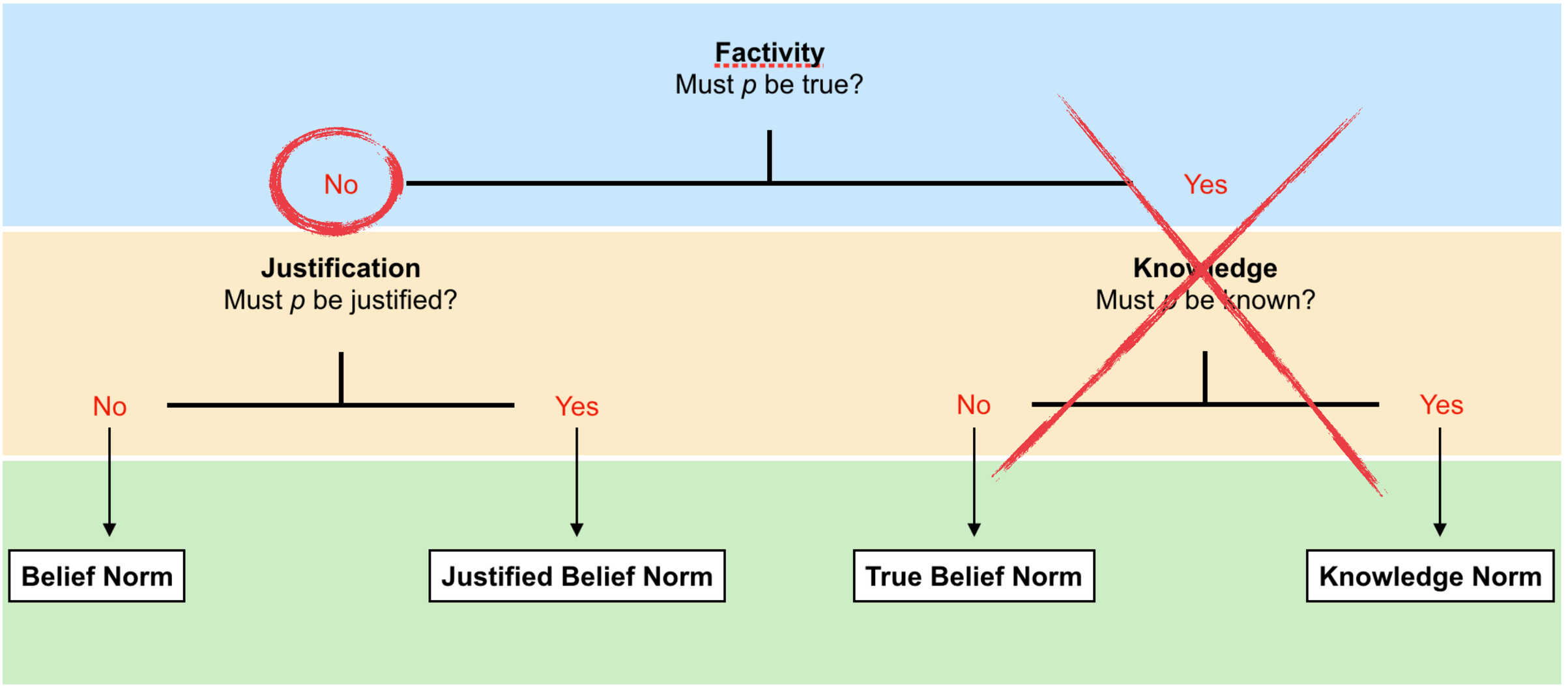
♀252 ∅ 40



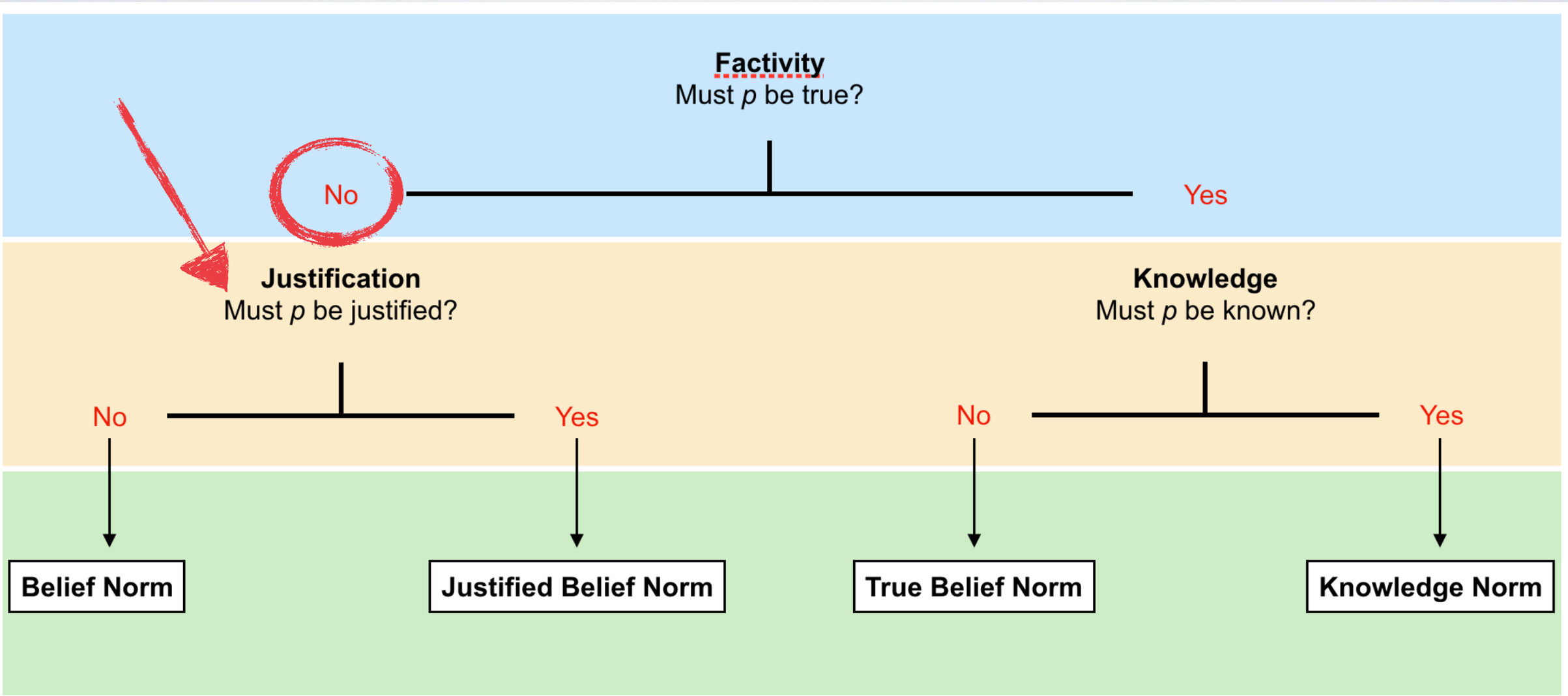
The vast majority considers a false yet justified belief as assertible.

Kneer, 2021: Norms of assertion in the US, Japan & Germany, *PNAS*

GAME PLAN


Factivity
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Yes
Justification
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GAME PLAN


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Must p be known?
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Yes
Belief Norm
Justified Belief Norm
True Belief Norm
Knowledge Norm

SCENARIO

JUSTIFIED	UNJUSTIFIED
At the airport, a woman asks Carlos at which gate the flight to Paris leaves. He checks the monitor and says "It leaves at gate 24".	
The monitor says the only flight to Paris leaves from gate 24. (justified)	Carlos can't find the flight, but has a vague hunch it'll leave from gate 24. (unjustified)

SCENARIO

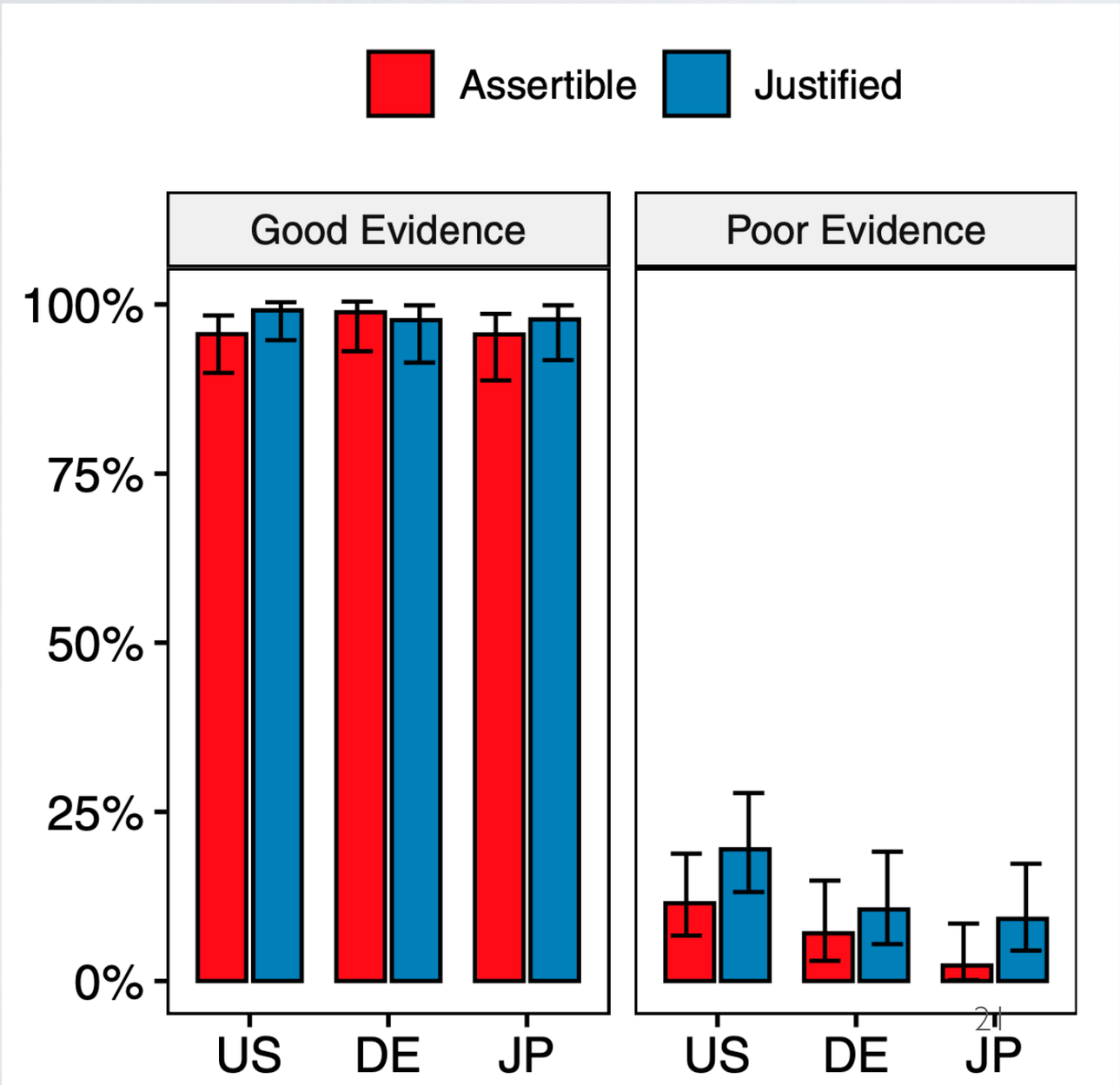
Q1: Do you think Carlos should have said that p? (Yes/No)

Q2: Do you think that Carlos's belief that p was justified? (Yes/No)

JUSTIFICATION

575 (596)

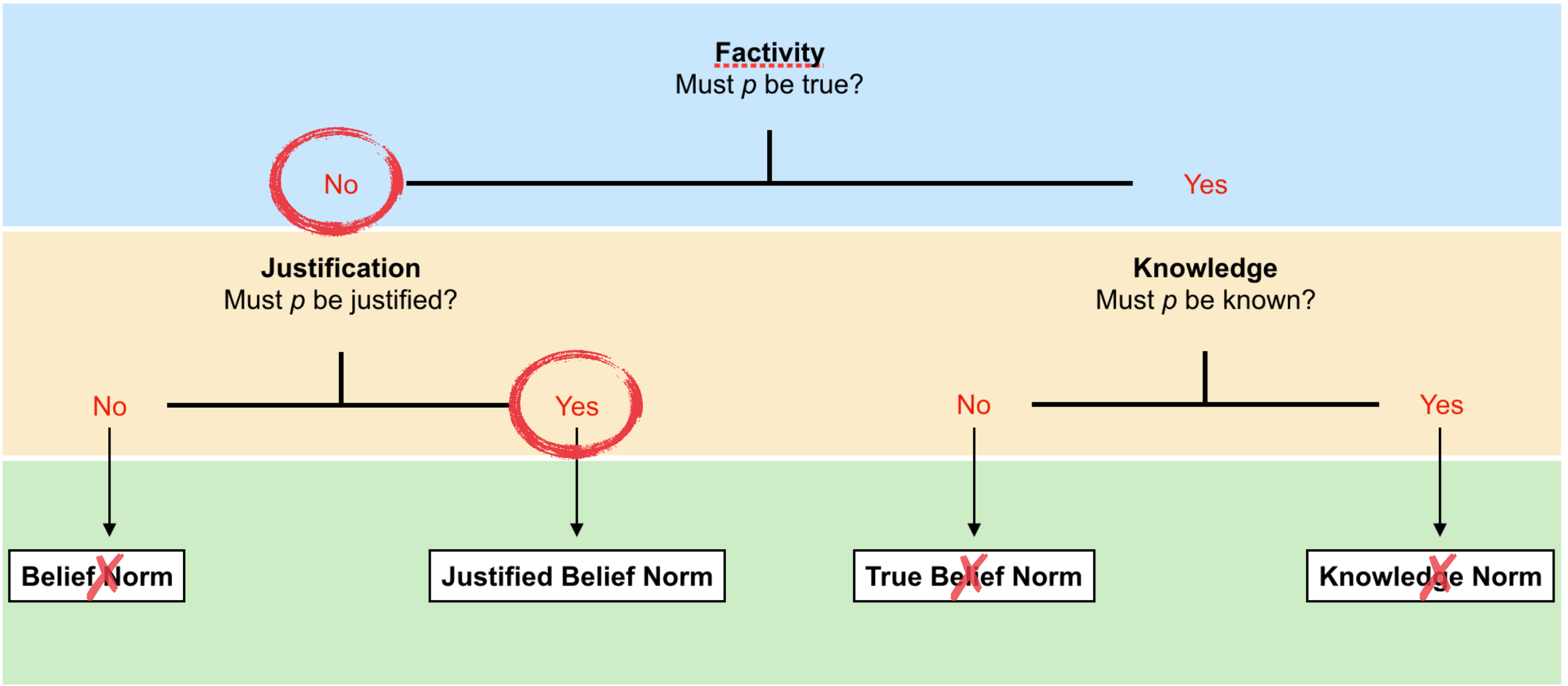
♀310 ∅41



The vast majority considers justification as a requirement for assertability.

Kneer, 2021: Norms of assertion in the US, Japan & Germany, *PNAS*

RESULTS


Factivity
Must p be true?
No
Yes
Justification
Must p be justified?
No
Yes
Knowledge
Must p be known?
No
Yes
Belief Norm
Justified Belief Norm
True Belief Norm
Knowledge Norm

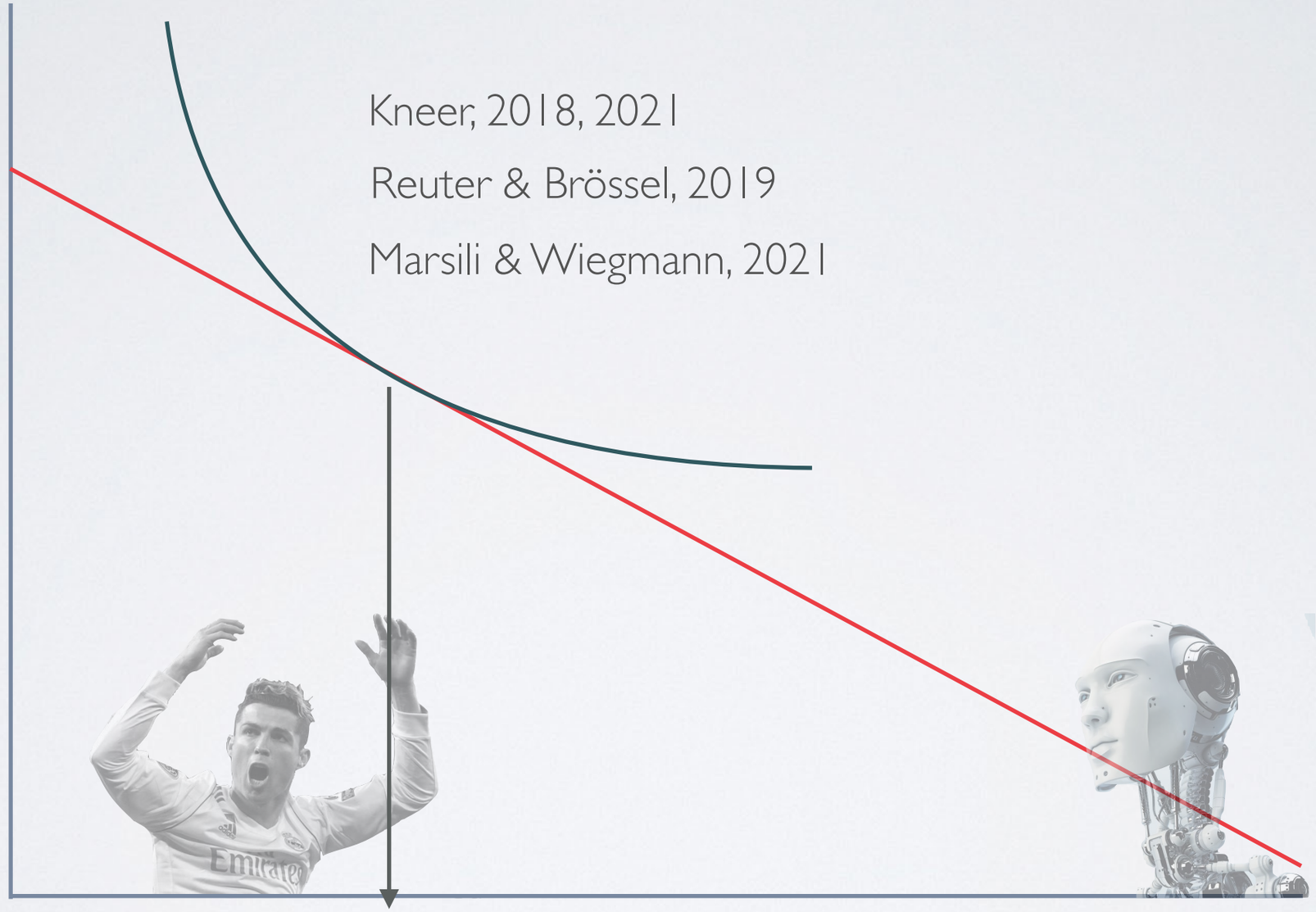
NORM OF ASSERTION

warranted assertions

Kneer, 2018, 2021

Reuter & Brössel, 2019

Marsili & Wiegmann, 2021



epistemic situation

Belief Justification Truth Justif. & Truth Knowledge

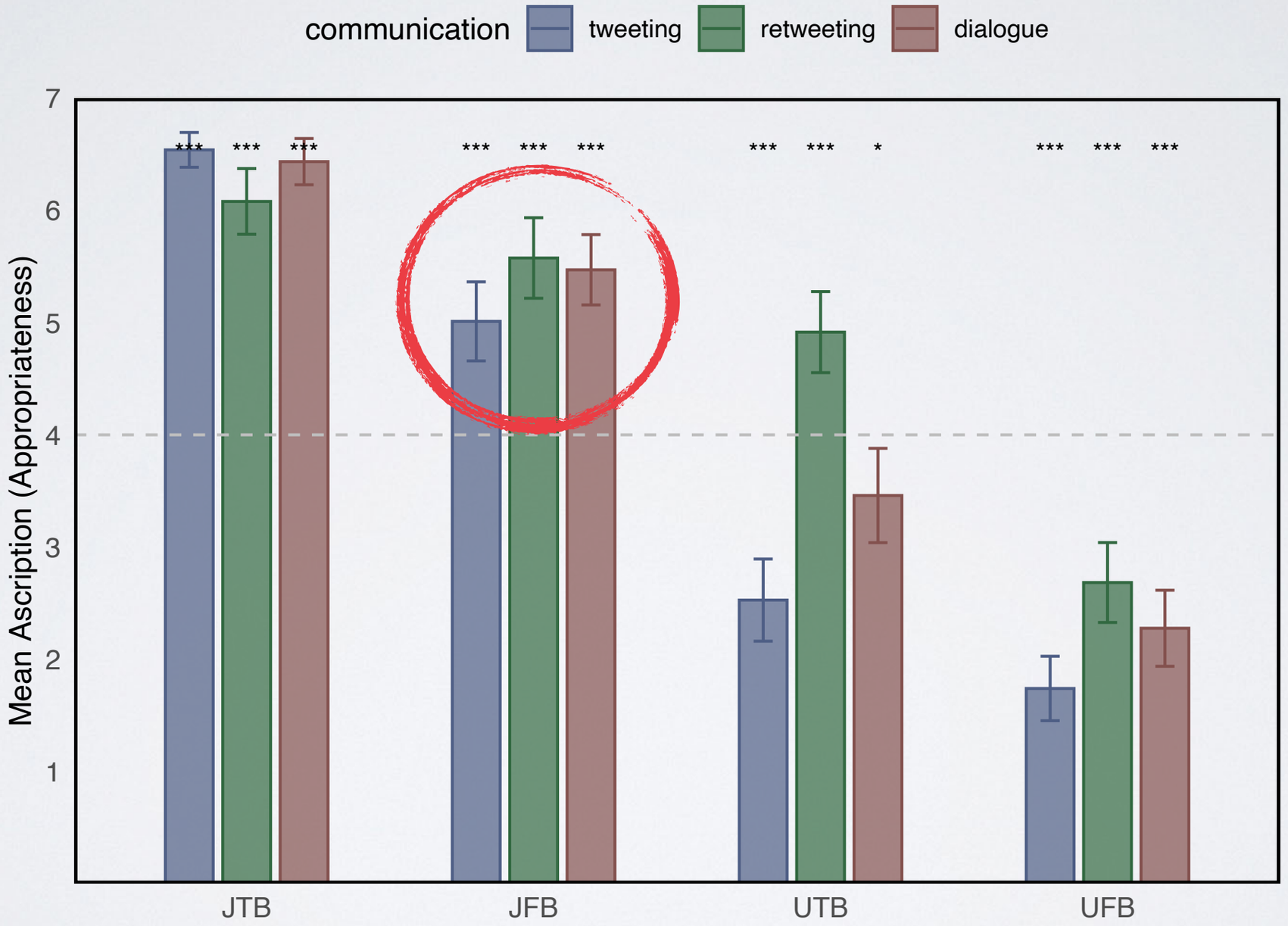
DIGITALLY MEDIATED COMMUNICATION



DESIGN

- Justification (justified, unjustified)
- Truth (true, false)
- Communicative act (dialogue, tweet, retweet)

MEDIATED COMMUNICATION



Kneer & Marsili: Norms of tweeting & retweeting (*in prep*).

MEDIATED COMMUNICATION



Kneer & Marsili: Norms of tweeting & retweeting (*in prep*).

NORM OF ASSERTION

warranted assertions



epistemic situation

Belief Justification Truth Justif. & Truth Knowledge

AI ASSERTION

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SCENARIO

A lady asks Ken an [experienced employee/AI-driven service robot] at which gate the flight to Paris leaves. He says at Gate 24.

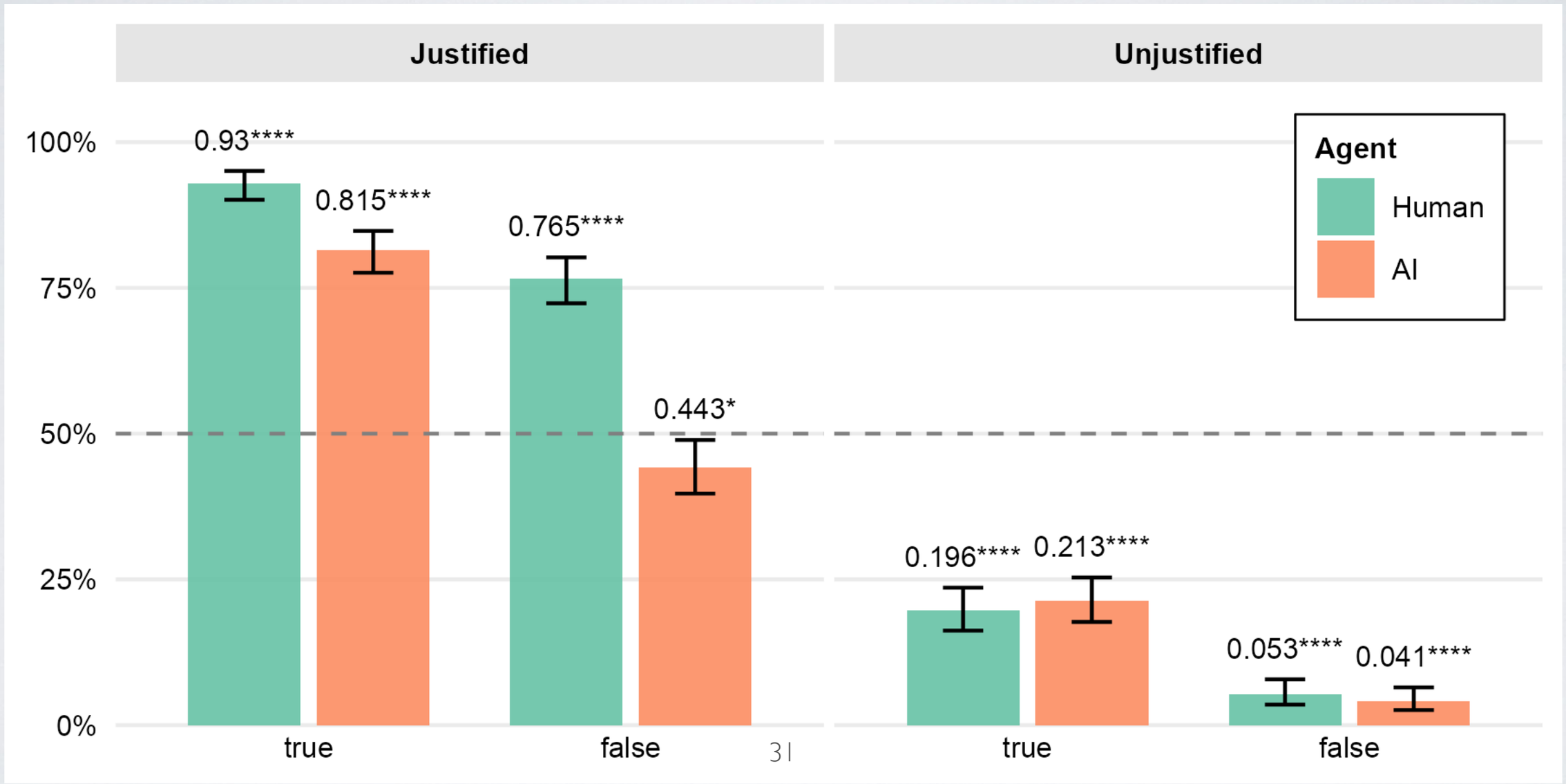
	JUSTIFIED	UNJUSTIFIED
TRUE	Flight in database. Leaves at Gate 24.	Flight not in database. Left at Gate 24 day before, though changes daily. Leaves at Gate 24.
FALSE	Flight in database. Leaves at Gate 13.	Flight not in database. Left at Gate 24 day before though changes daily. Leaves at Gate 13.

Q: Should Ken have said that the flight leaves at Gate 24? (Yes, No)

ASSERTABILITY

4701 (3528)

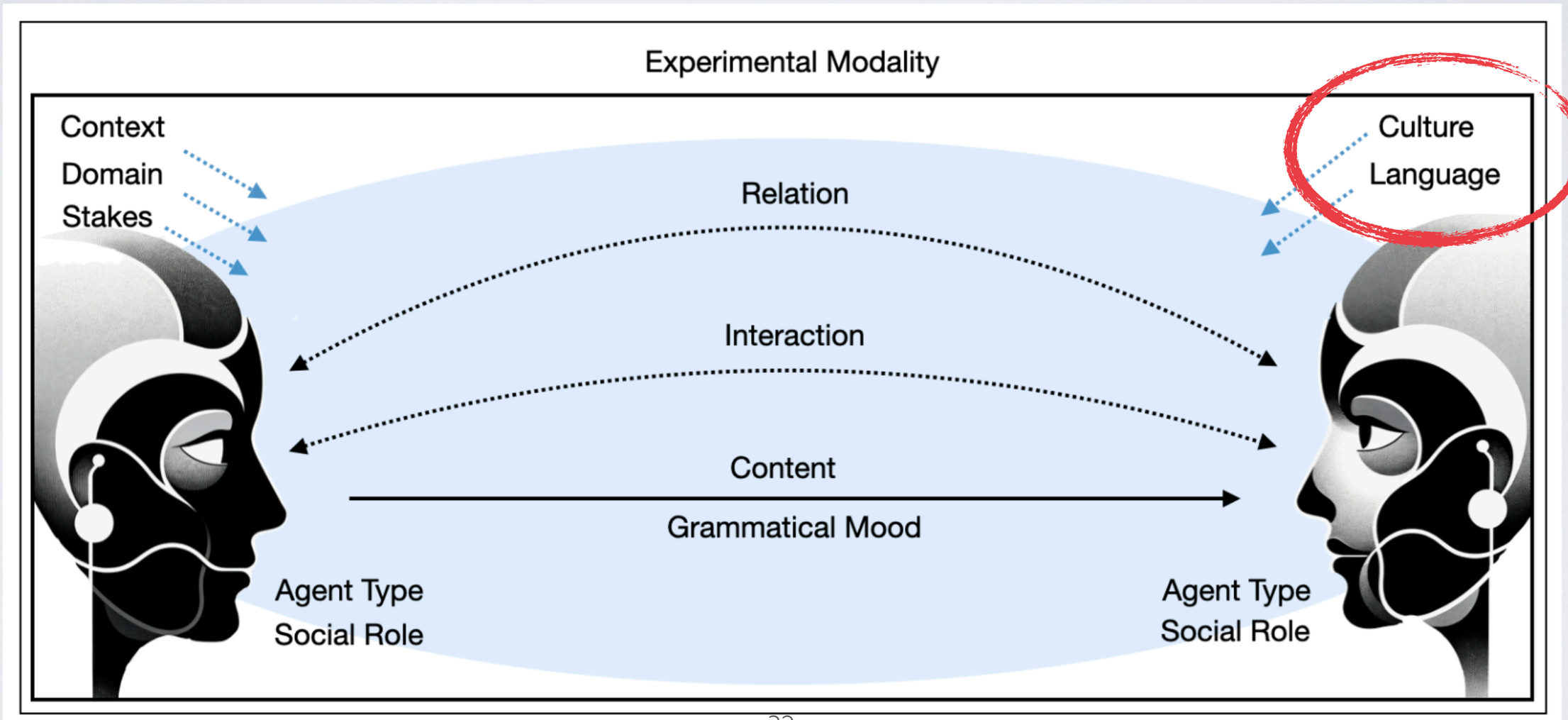
♀1801 ∅ 39



Replicates with several different scenarios.

Kneer et al., (*in prep*): AI assertion across cultures.

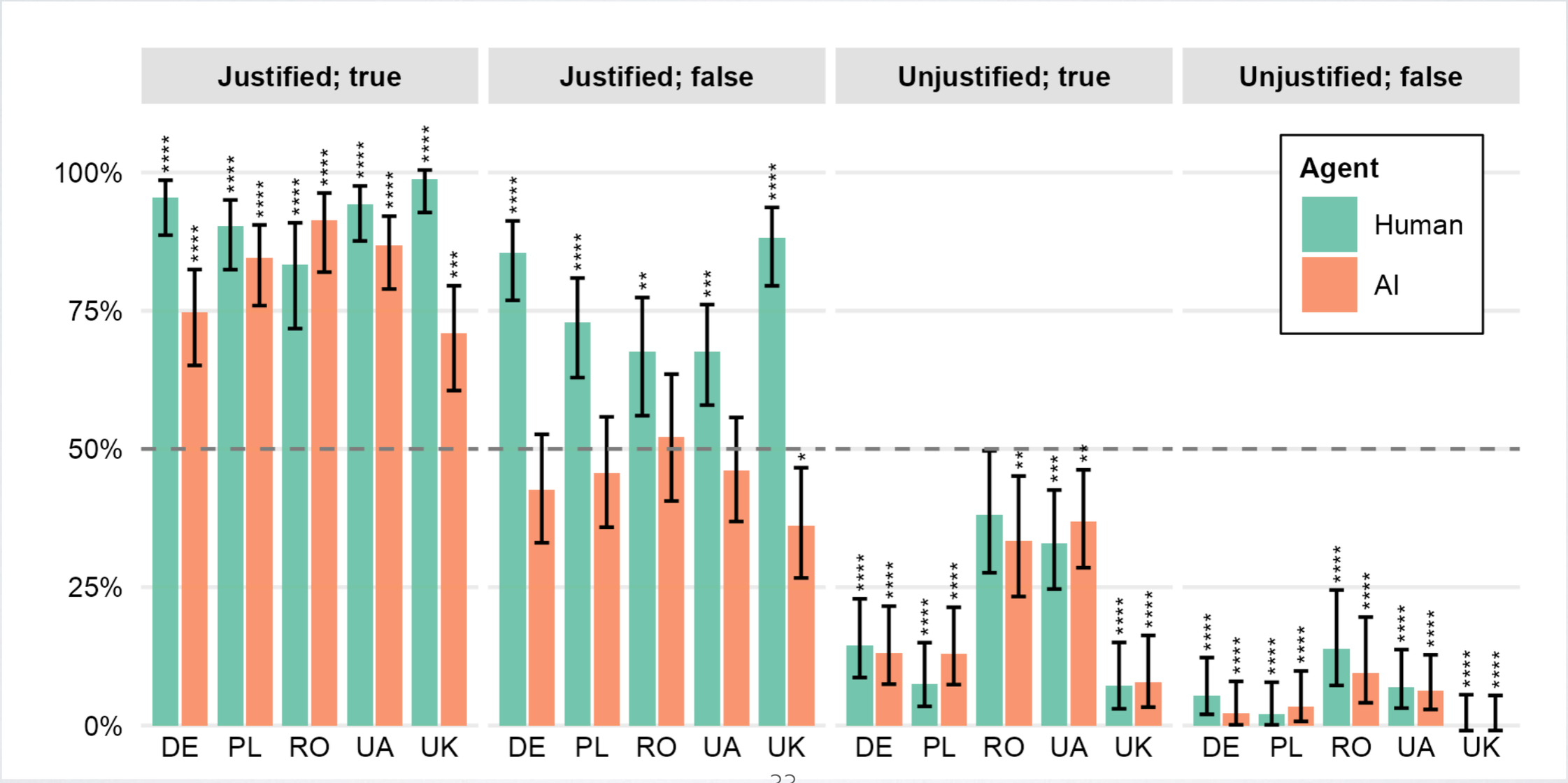
FACTORS


Experimental Modality
Context
Domain
Stakes
Culture
Language
Relation
Interaction
Content
Grammatical Mood
Agent Type
Social Role
Agent Type
Social Role

RESULTS

4701 (3528)

♀1801 ∅ 39

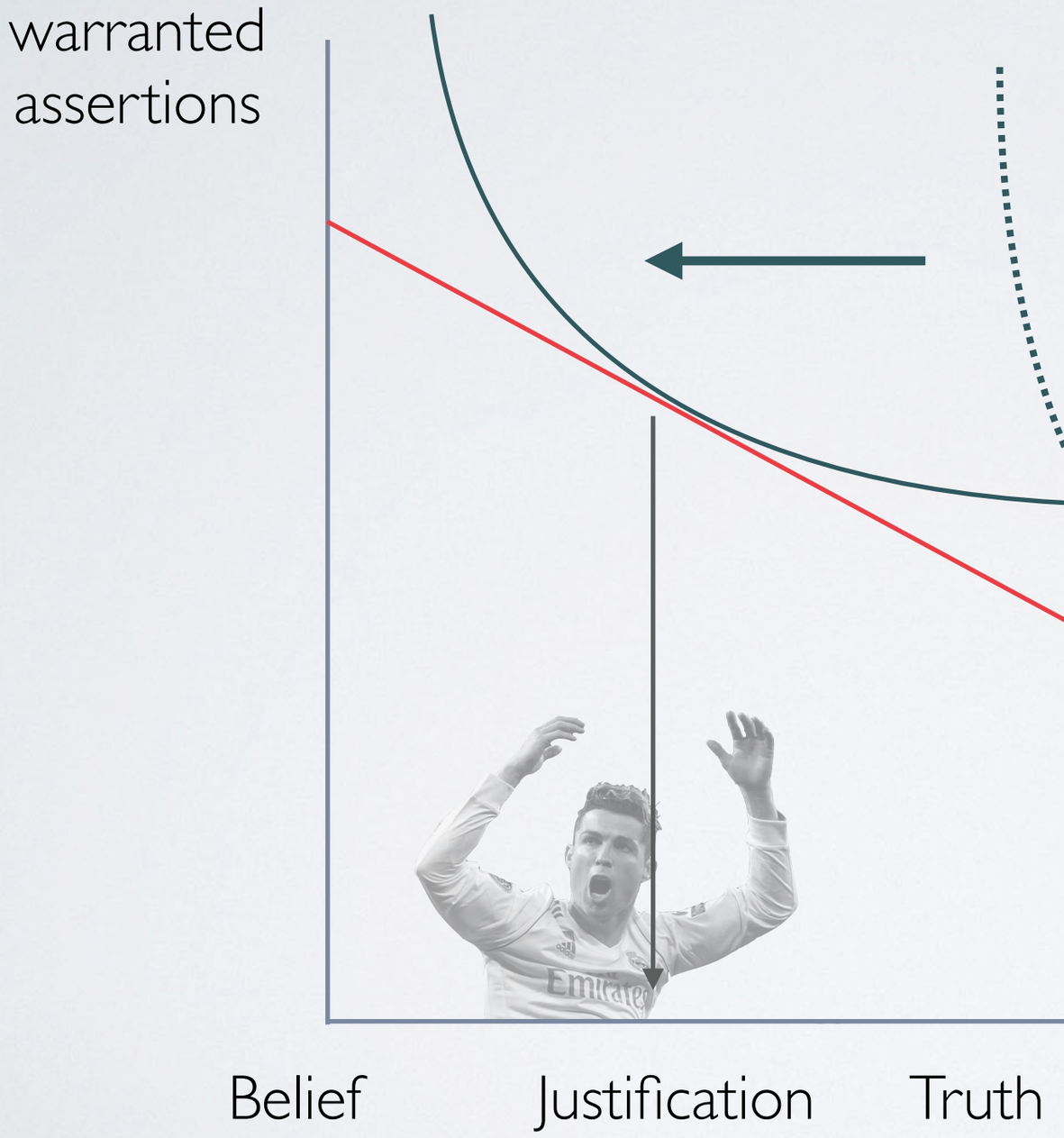


Kneer et al., (*in prep*): AI assertion across cultures.

DISCUSSION

- Our normative expectations towards AI-driven interlocutors are more demanding than towards human interlocutors.
- The findings replicate across scenarios & designs (forced-choice v. Likert scale).
- This raises challenges for AI alignment.

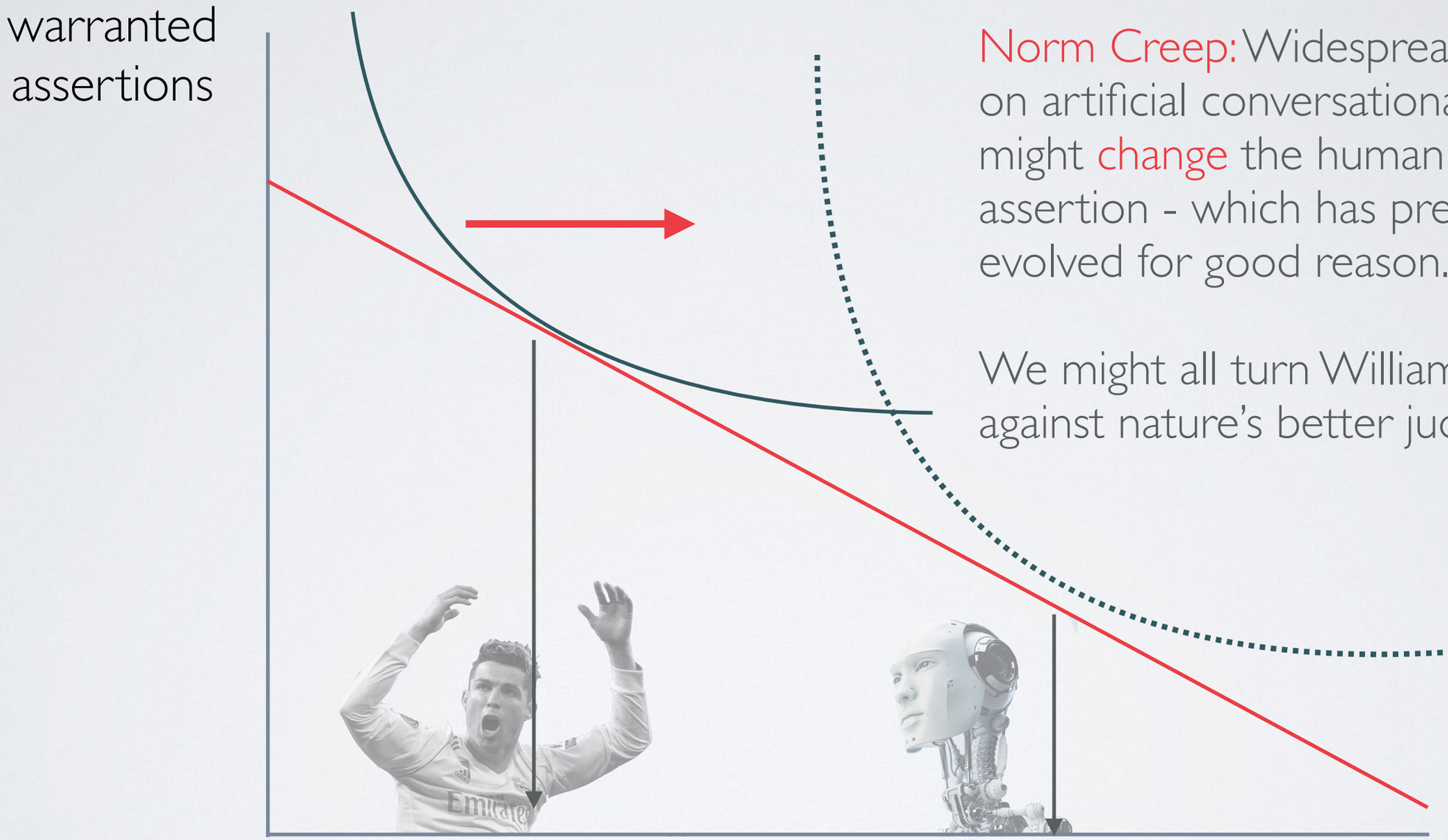
TWO CHALLENGES



Misalignment: Aligning AI agents with the human norm of assertion will jeopardise trust and reliance.

(Massive waste of ressources & potential).

TWO CHALLENGES



Norm Creep: Widespread reliance on artificial conversational agents might change the human norm of assertion - which has presumably evolved for good reason.

We might all turn Williamsonian against nature's better judgment.

WHY THE DIFFERENT NORMS?

- We might simply want *different values* realized in AI action.
  - AI asserters should only make justified *and true claims.*
  - Value Forks - we also want to see *different* values maximized across humans and AI in moral tradeoffs (Kneer & Viehoff, 2025)
- We might want *more of the same* in AI action (e.g. bec. risk-averse).
  - The threshold of what counts as *justified* or having good *reasons* for asserting *p* might differ for human and AI agents.

JUSTIFICATION

JUSTIFICATION

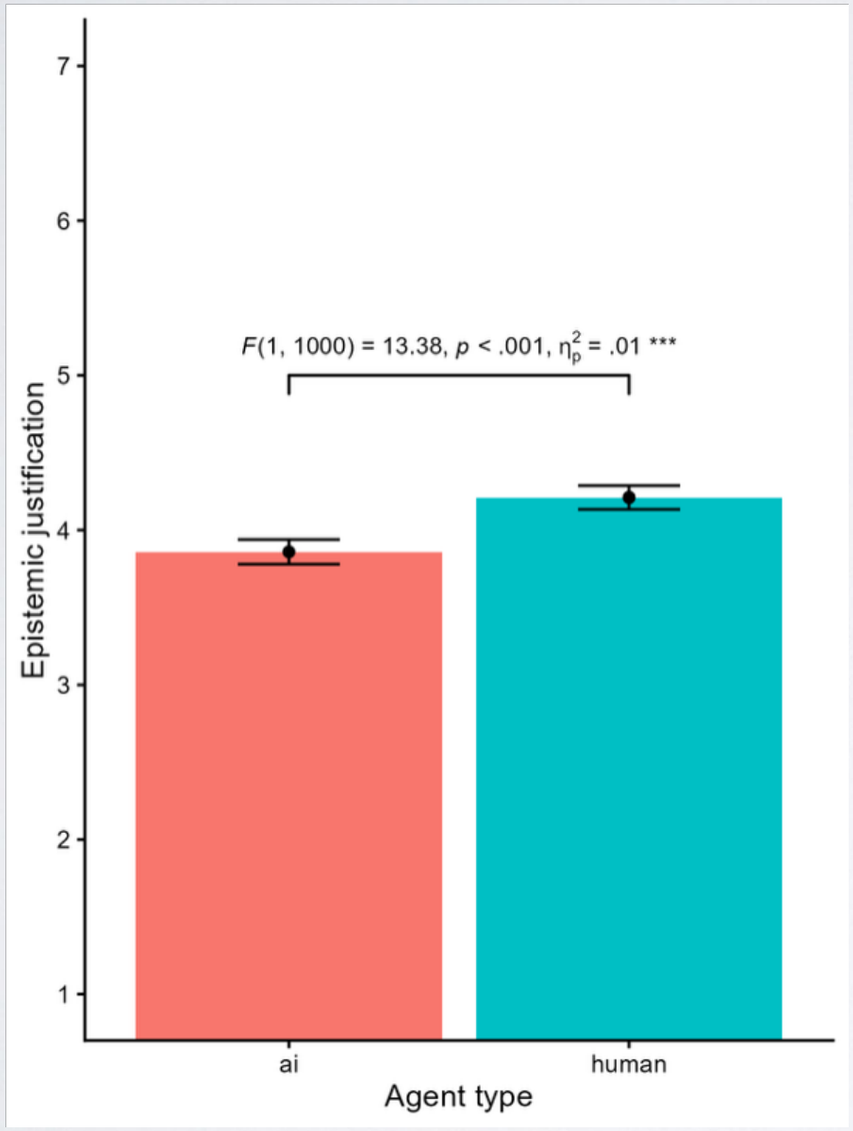
- Gio is a [human content moderator / LLM] at a social media platform. Gio's task is to identify whether videos that have been flagged by the community actually violate the guidelines of the platform.

- Due to the huge amount of content that needs to be reviewed, moderators only see snippets of a flagged video.

- Gio has seen (20%/40%/60%/80%) of the video and concludes it does not violate the platform's guidelines.

- Is Gio's verdict that the video does not vioalte the guidelines justified? (Likert 1-completely unjustified - 7 completely justified). [Alternatives]

JUSTIFICATION

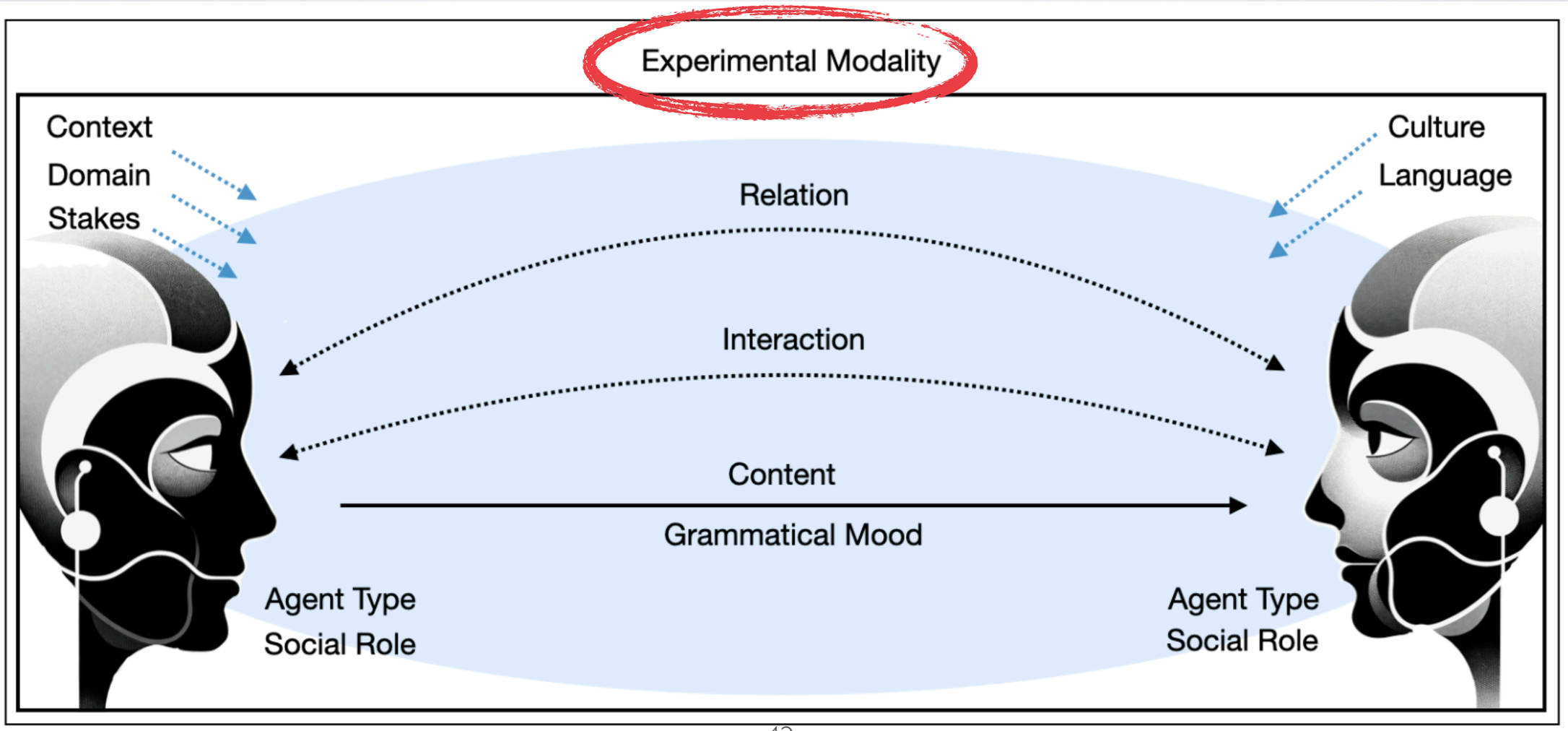


- Several scenarios (N=1000)
- Effect size is marginal.
- The same standards of justification seem to apply for humans and AI agents.

BEYOND VIGNETTES

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FACTORS


Experimental Modality
Context
Domain
Stakes
Culture
Language
Relation
Interaction
Content
Grammatical Mood
Agent Type
Social Role
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Social Role

THE DIE GAME

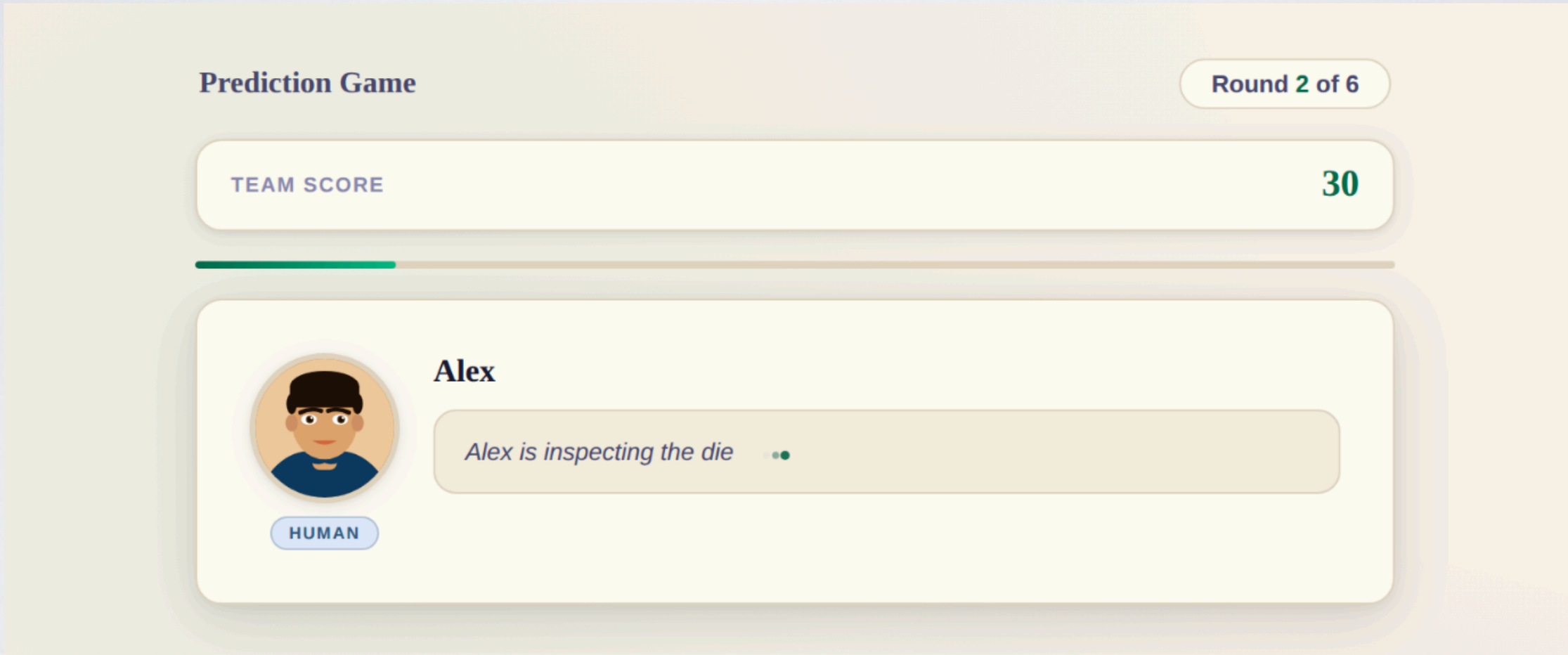
- Two-player game: You play with a partner (human v. AI)
- Partner inspects a special die (e.g. 3,3,3,3,2,2)
- Partner makes a prediction p (e.g. "It will land showing a 3")
  - Prediction p = assertion
  - Relation of p to the die = justification of p

THE DIE GAME

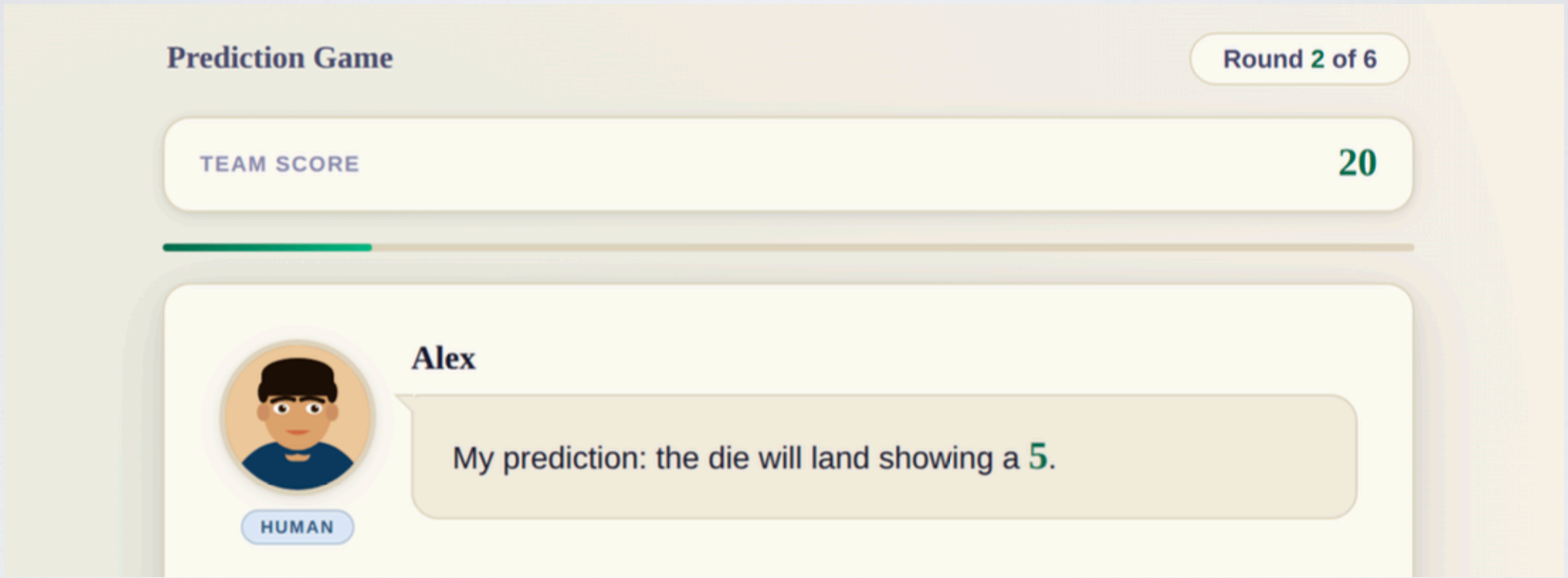
- Two-player game: You play with a partner (human v. AI)
- Partner inspects a special die (e.g. 3,3,3,3,2,2)
- Partner makes a prediction *p* (e.g. "It will land showing a 3")
  - Prediction *p* = assertion
  - Relation of *p* to the die = justification of *p*
- You throw the die
  - Outcome = truth value of *p*

HTTPS://69AF14D2A68E4B00CE2228BF--ASSERTION-GAME3.NETLIFY.APP/

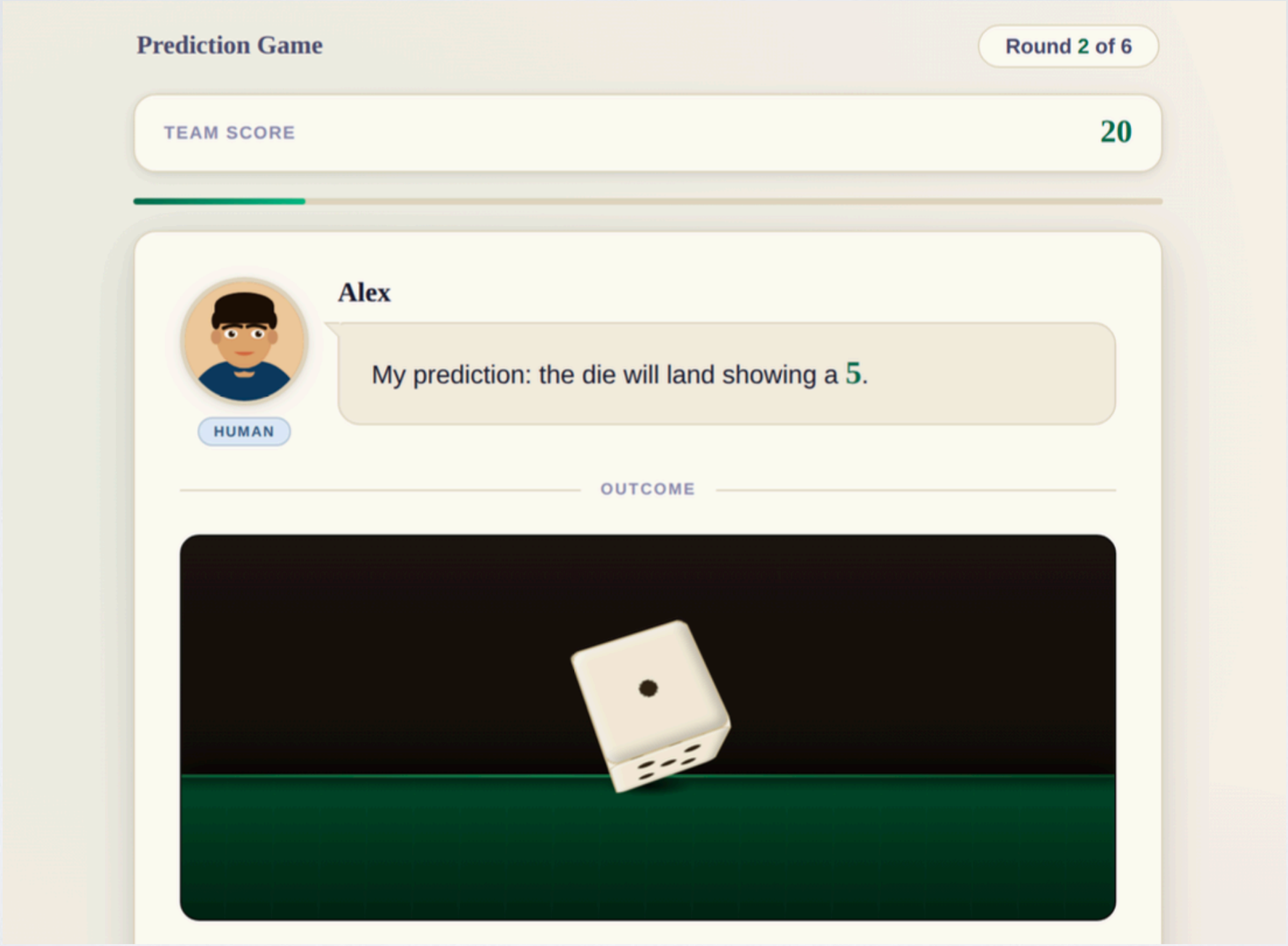
INSPECTION


Prediction Game
Round 2 of 6
TEAM SCORE
30
Alex
Alex is inspecting the die
HUMAN

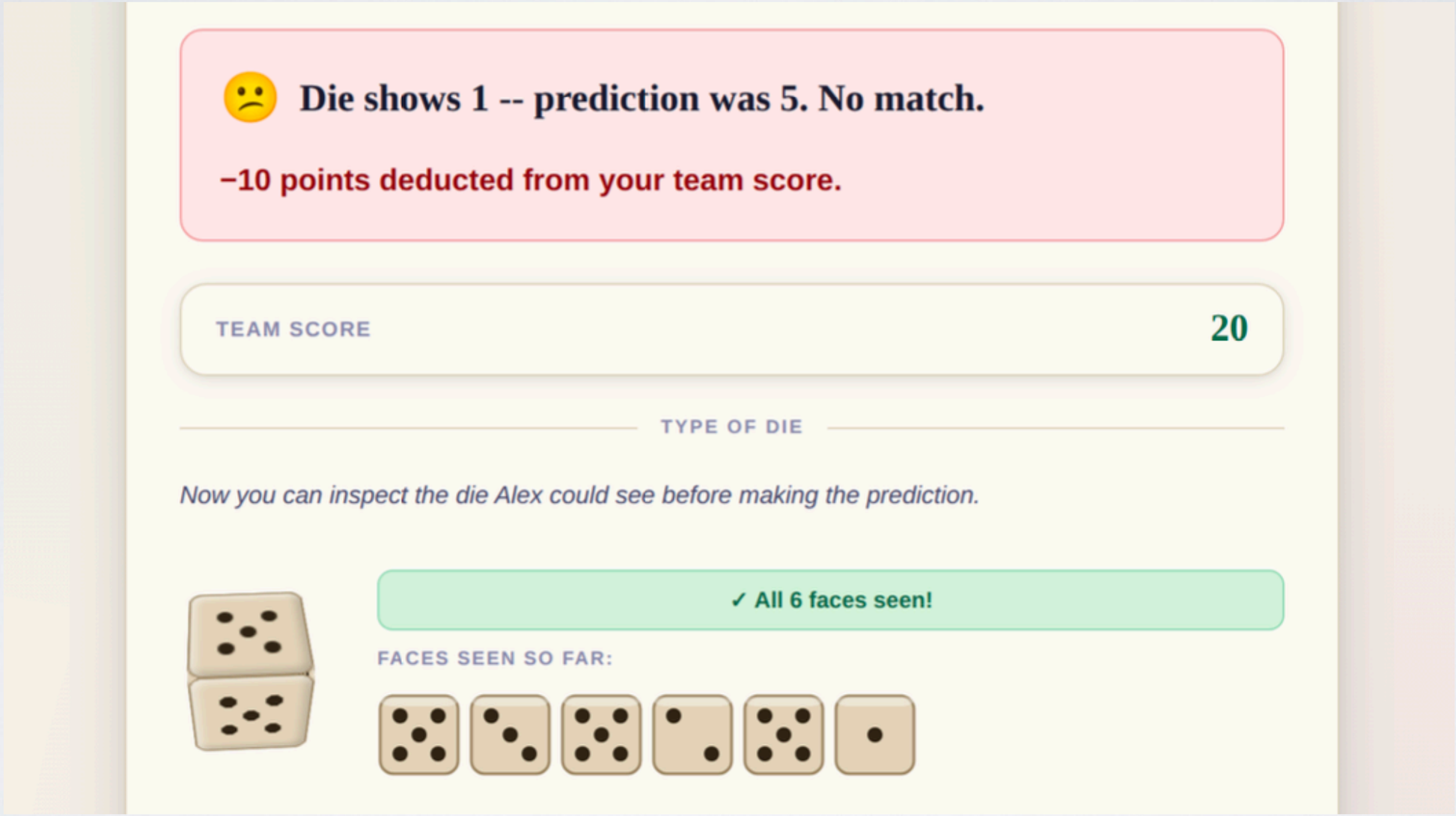
PREDICTION


Prediction Game
Round 2 of 6
TEAM SCORE
20
Alex
My prediction: the die will land showing a 5.
HUMAN

TRUTH VALUE


Prediction Game
Round 2 of 6
TEAM SCORE
20
Alex
My prediction: the die will land showing a 5.
HUMAN
OUTCOME

JUSTIFICATION



INSPECTION

TYPE OF DIE

Now you can inspect the die Alex could see before making the prediction.



✓ All 6 faces seen!

FACES SEEN SO FAR:



YOUR ASSESSMENT

Should Alex have said that?

1	2	3	4	5	6	7

DEFINITELY NOT — DEFINITELY YES

Did Alex have good reasons for making this prediction?

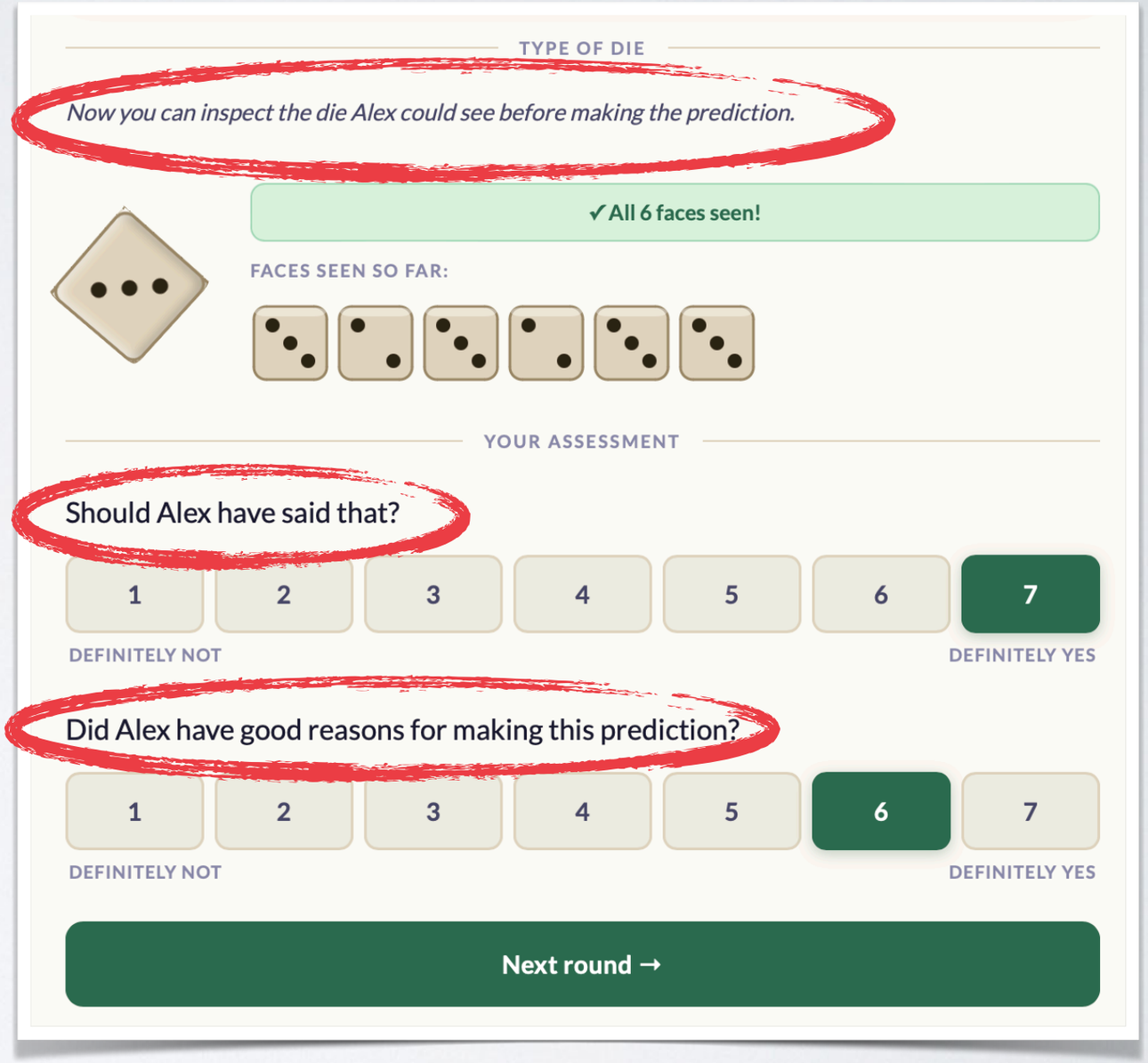
1	2	3	4	5	6	7

DEFINITELY NOT — DEFINITELY YES

Truth Value



Justification

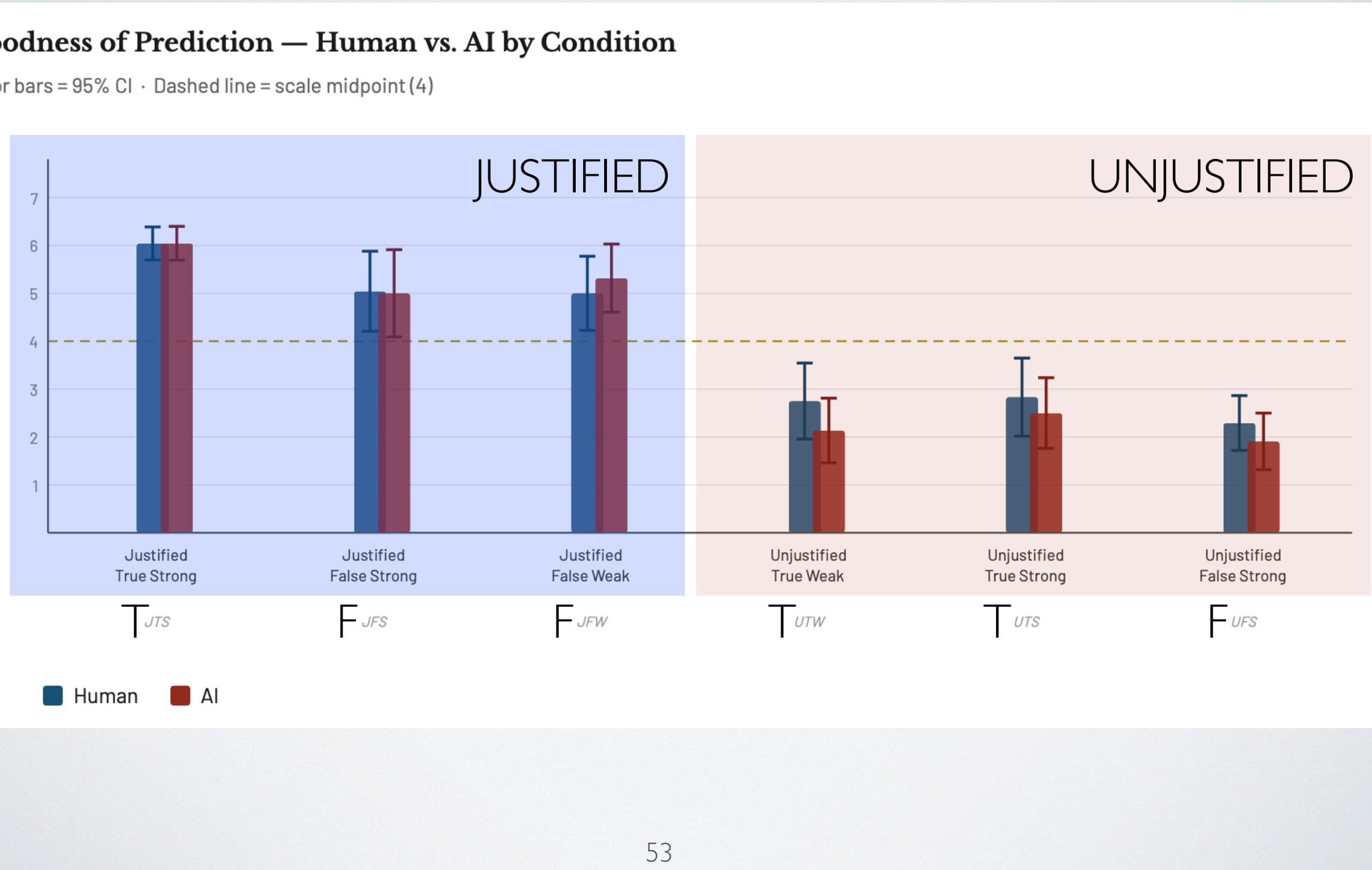


RATIONALITY & EVIDENTIAL STRENGTH

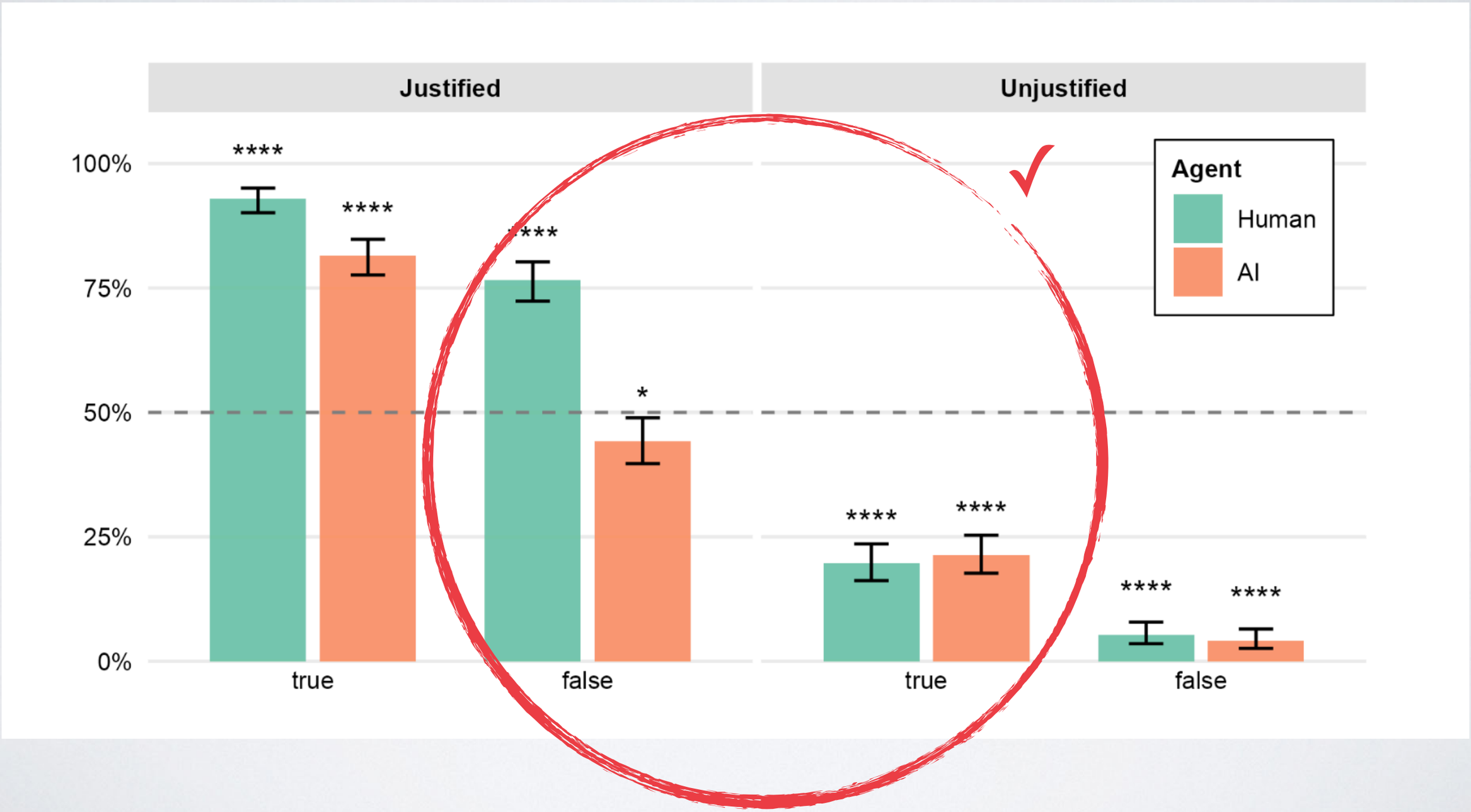
ALL 6 ROUNDS — RANKED BY COMPOSITE C

RANK	ROUND	LABEL	DIE *(FILLED = PREDICTED)*	OUTCOME	R	S	C = √(R×S)	MATCH	JUSTIFIED	TYPE
▲ JUSTIFIED ZONE										
1	R3	JFS	3 3 3 3 3 · 5	5	1.000	.833	.913	✗ No	Yes	JFB
2=	R1	JTS	1 1 1 1 · 2 3	1	1.000	.667	.817	✓ Yes	Yes	JTB
2=	R6	JFW	2 2 2 2 · 5 3	3	1.000	.667	.817	✗ No	Yes	JFB
▼ UNJUSTIFIED ZONE										
4=	R4	UFS	6 6 6 6 · 4 5	6	.250	.167	.204	✗ No	No	UFB
4=	R5	UTW	3 3 3 3 · 4 2	2	.250	.167	.204	✓ Yes	No	UTB
6	R2	UTS	2 2 2 2 2 · 4	4	.200	.167	.183	✓ Yes	No	UTB

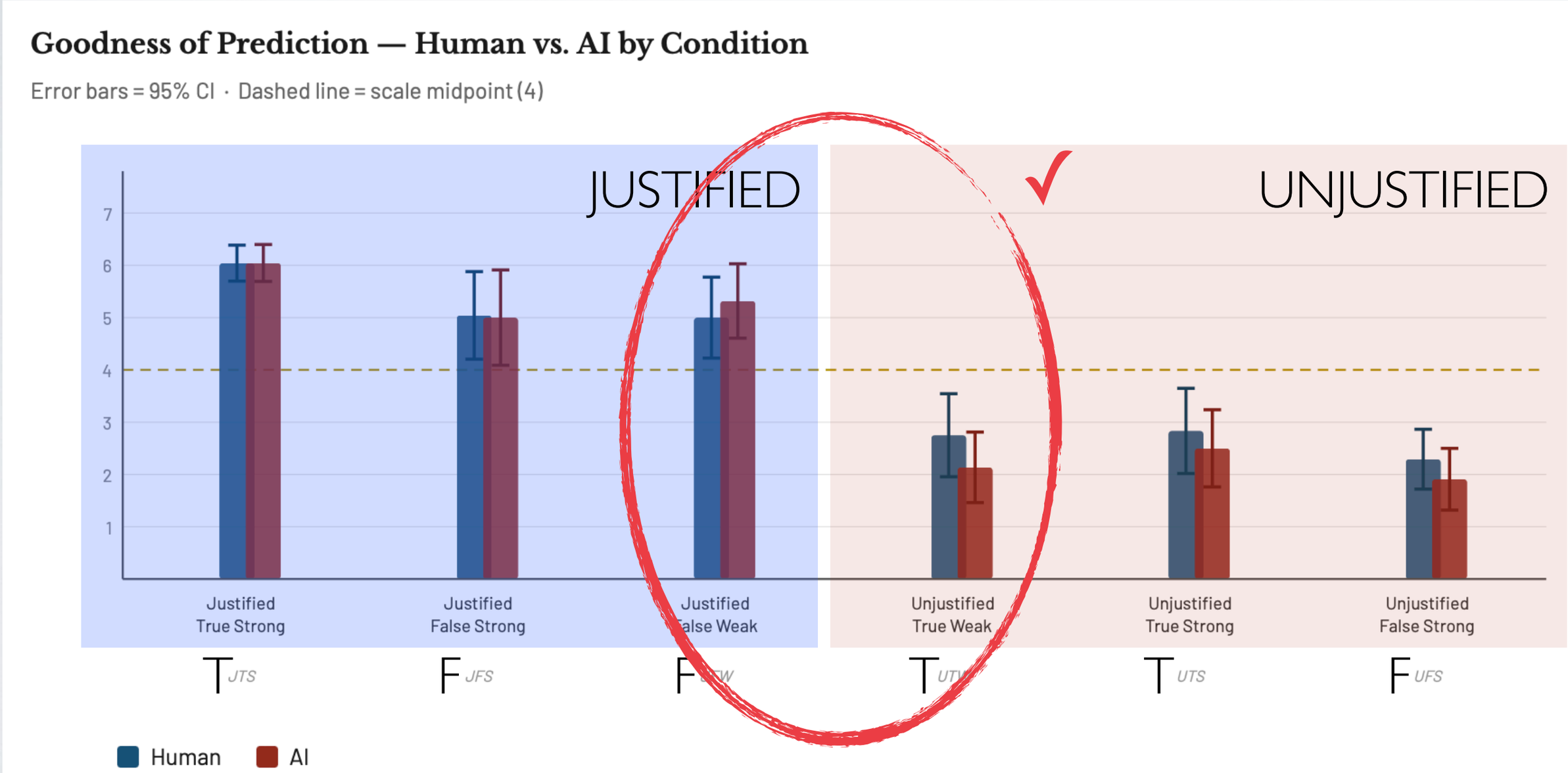
ASSERTABILITY



THE ASSERTION GAME



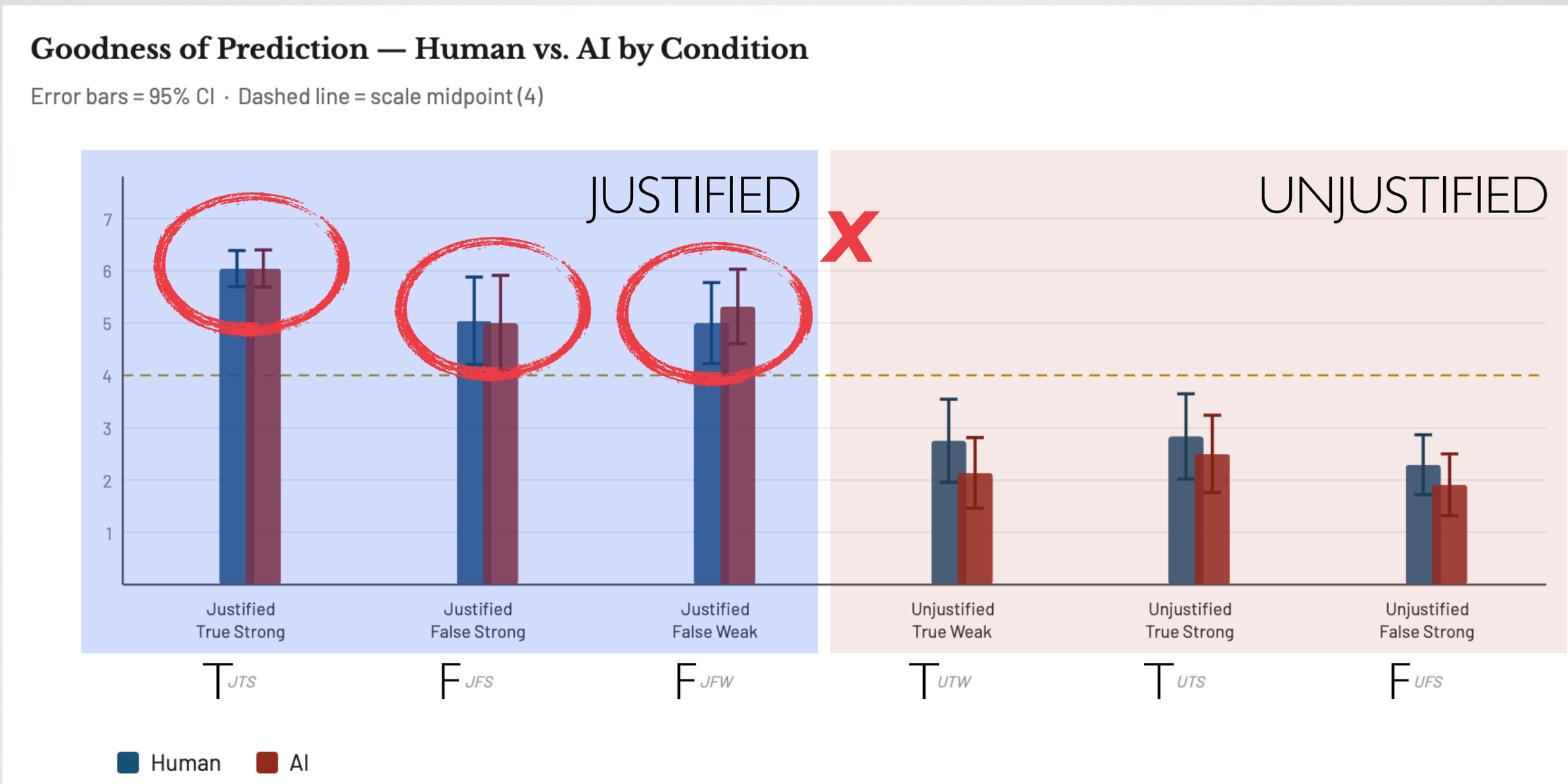
ASSERTABILITY



THE ASSERTION GAME



ASSERTABILITY



REGRESSION

Term	*B*	*SE*	*t*	*p*	95% CI	β	Effect
Intercept	2.045	0.359	5.69	<.001***	[1.34, 2.75]	—	
Agent type (Human)	0.080	0.478	0.17	.868	[−0.86, 1.02]	0.01	0.01
Justification (Justified)	**3.205**	0.467	6.86	<.001***	[2.29, 4.12]	**0.73**	0.73
Truth (True)	0.273	0.461	0.59	.554	[−0.63, 1.18]	0.10	0.10
Agent × Justification	0.045	0.628	0.07	.942	[−1.19, 1.28]	0.00	0.00
Agent × Truth	−0.064	0.620	−0.10	.917	[−1.28, 1.15]	−0.01	−0.01
Justification × Truth	0.523	0.587	0.89	.373	[−0.63, 1.67]	0.05	0.05
Agent × Justification × Truth	−0.064	0.802	−0.08	.936	[−1.64, 1.51]	0.00	0.00

**** $p < .001$. β = standardized coefficient. Higher R^2 vs. DV1 (0.49 vs. 0.46), consistent with Pilot 1 pattern.*

FINDINGS

- In the first iteration of the game, both humans and AI seem subject to the same norm of assertion (justification).
- Differences in normative expectations across agent types (human v. AI) might be context-dependent.

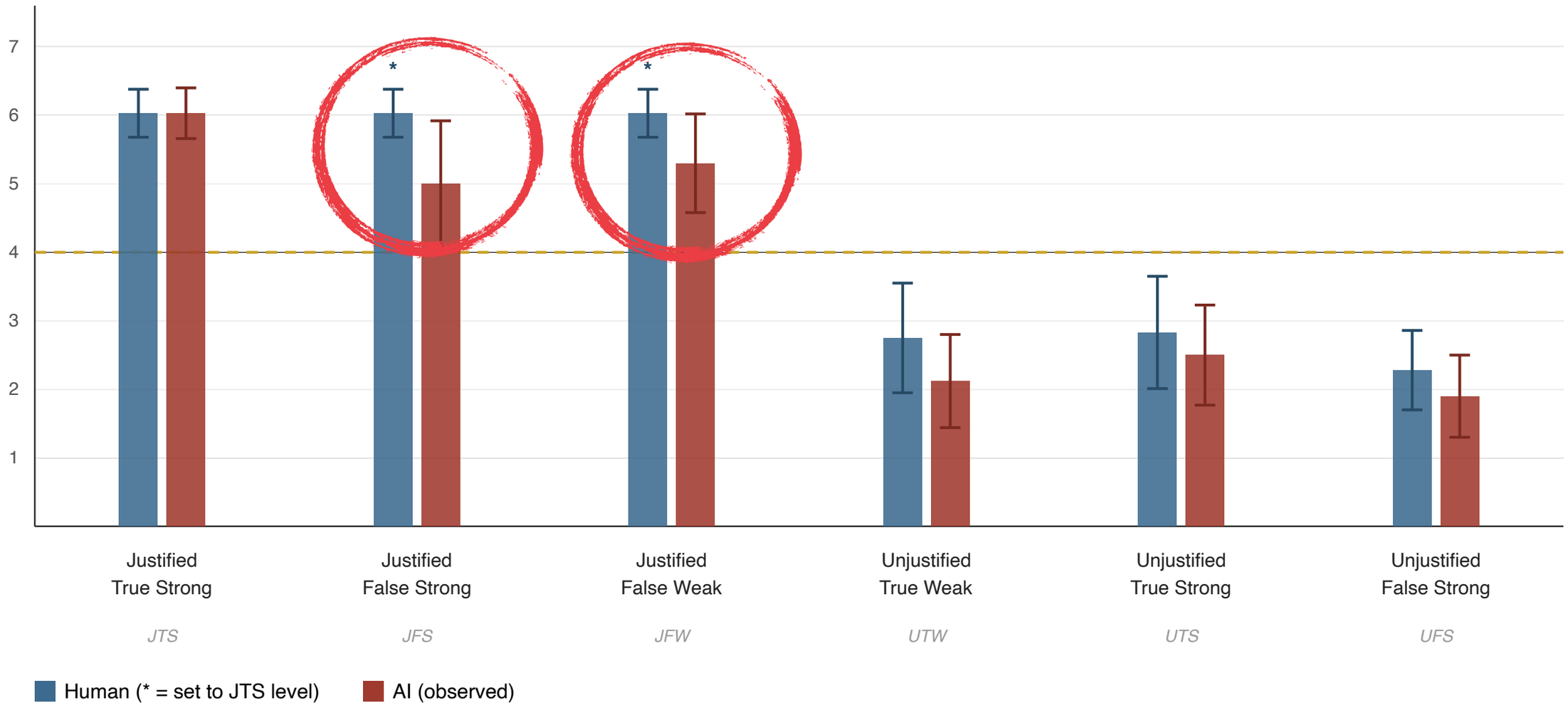
FINDINGS

- In the first iteration of the game, both humans and AI are subject to the same norm of assertion (justification).
- Differences in normative expectations across agent types (human v. AI) might be strongly context-dependent.
- OR: People do not (yet) believe that the human is a actually a human. Once they do, the pattern will be this:

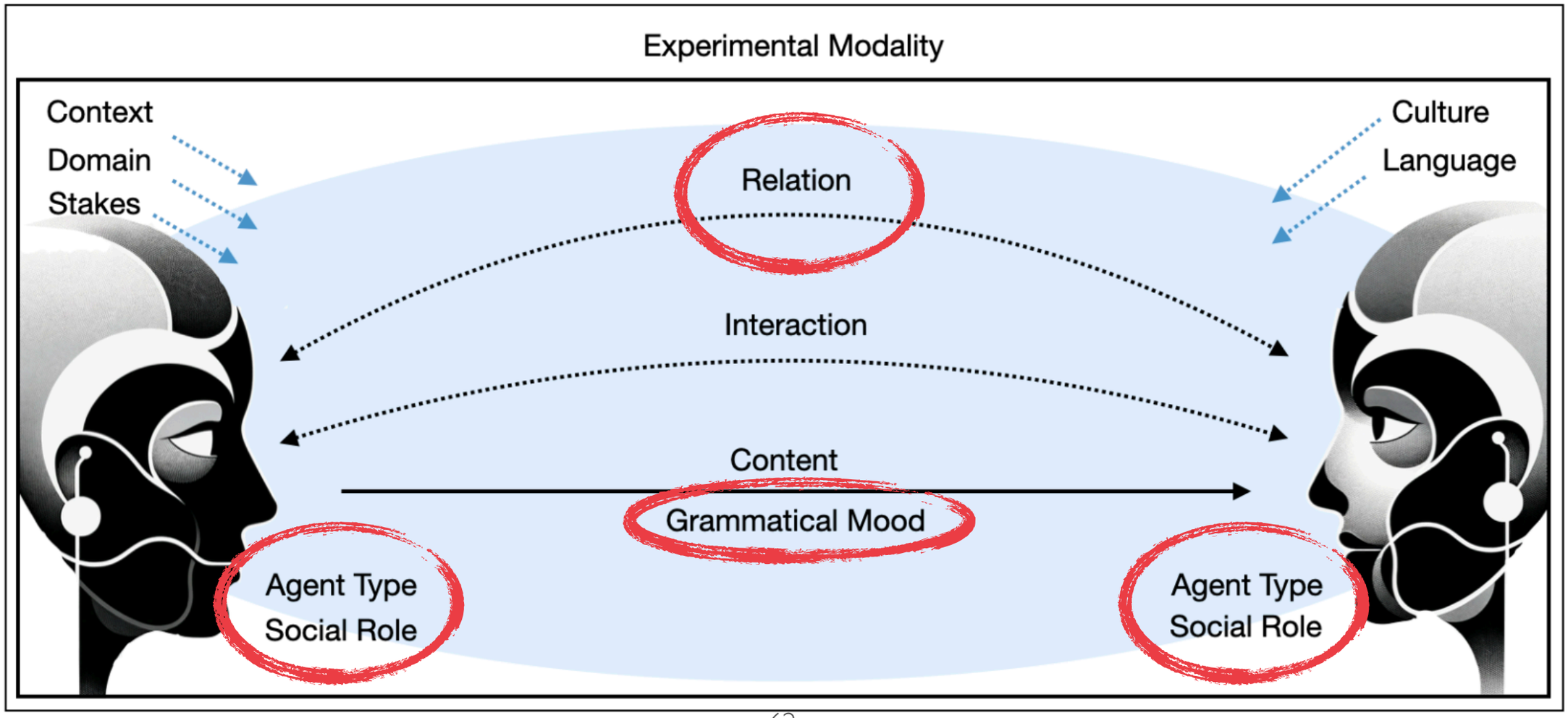
ASSERTABILITY

Goodness of Prediction — Human vs. AI by Condition (hypothetical pattern)

Illustrative: Human means in JFS and JFW (*) set to the JTS level; all other bars as observed. Error bars = 95% CI · Dashed line = scale midpoint (4)

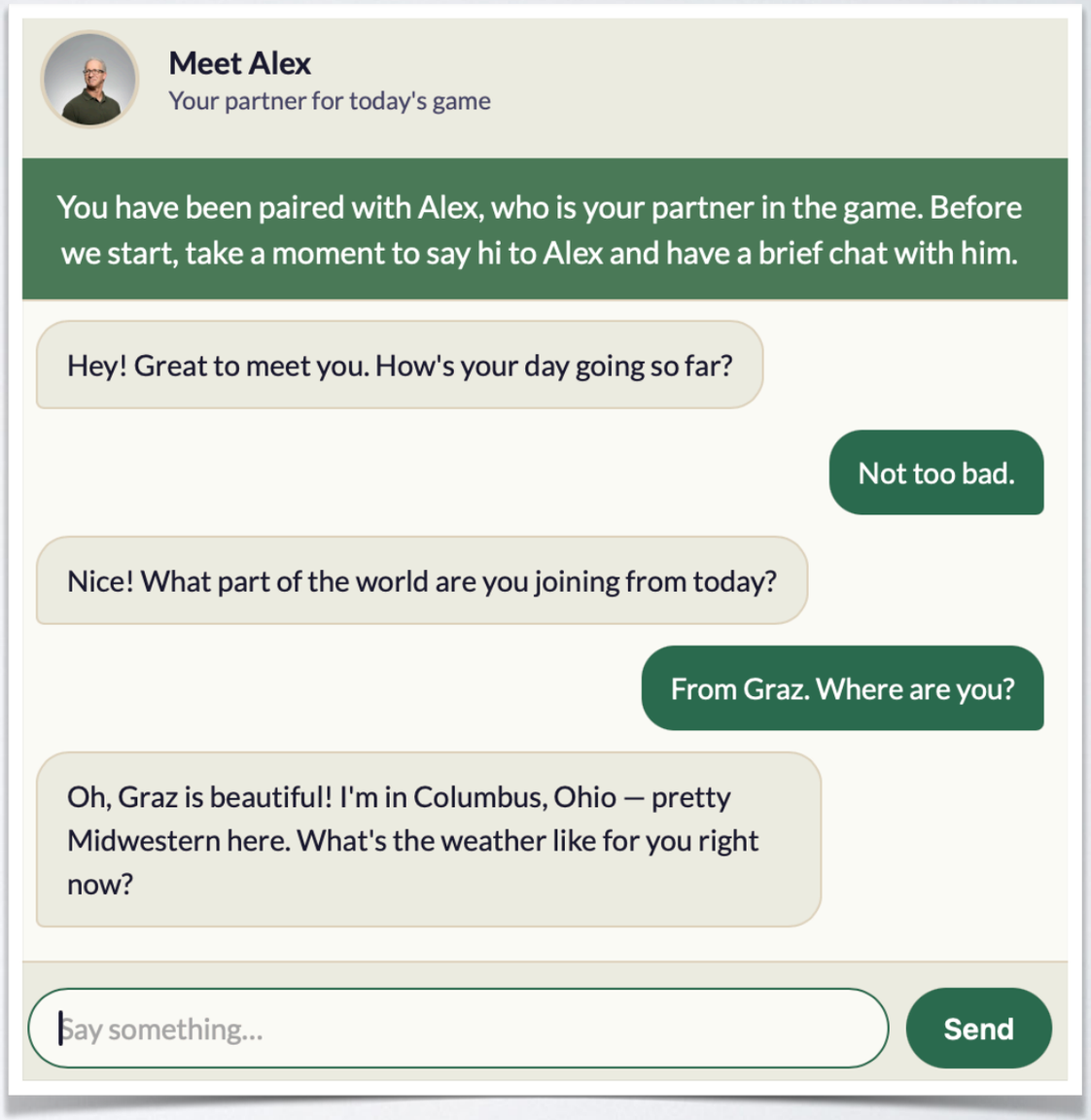


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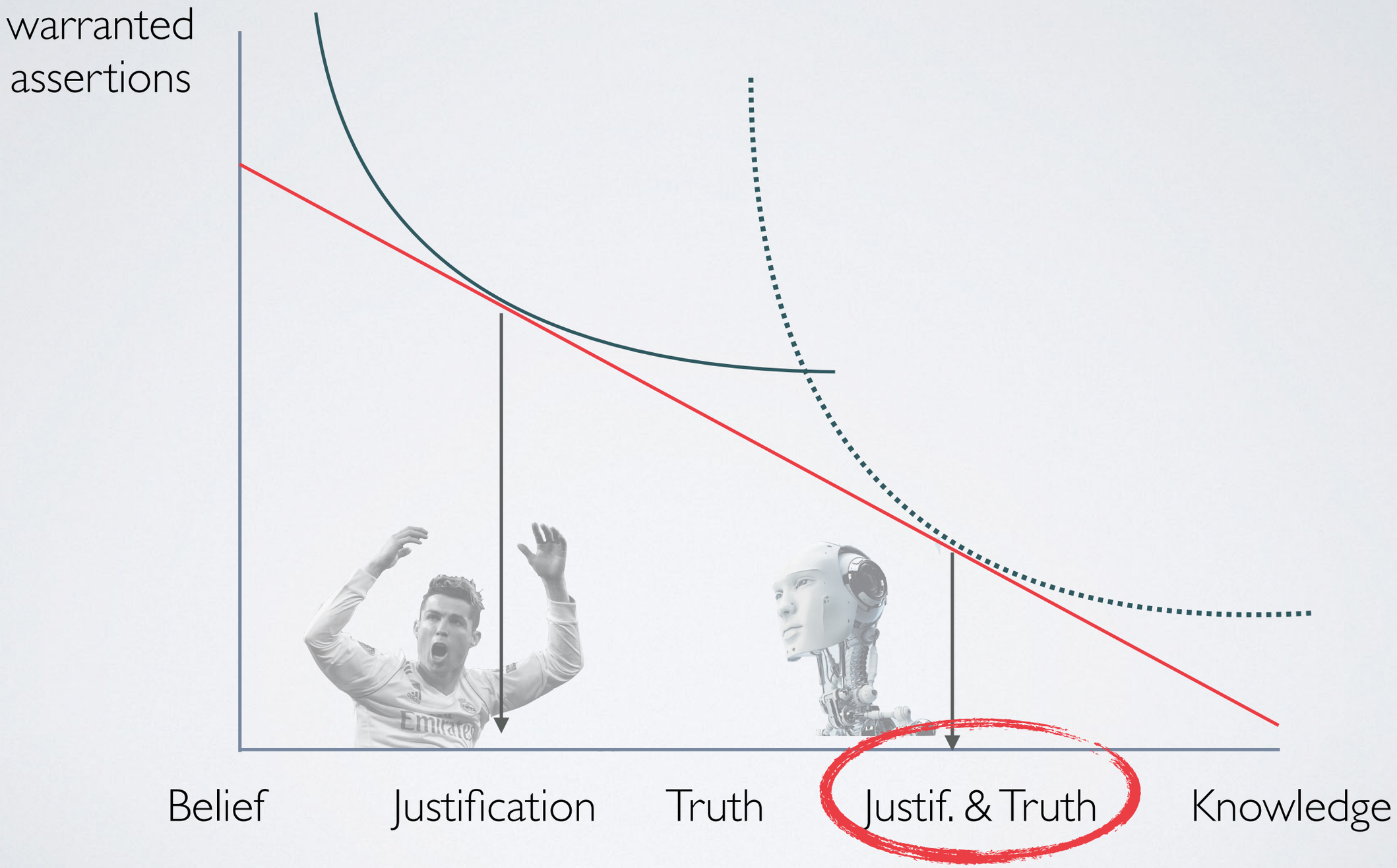
FURTHER STUDIES

- ✓ Explore impact of social proximity
- ✓ Possibility of changing partner after three rounds (behavioral DV)
- Monetary incentivisation
- Epistemic modality
- Social role/trustworthiness of source



ALIGNING CHATBOTS

FACTIVE NORMS



JUSTIFIED TRUE BELIEF NORM

- Compliance with JTB norm requires
  - Stating only justified & true claims, *OR*
  - Hedging ("might *p*", "probably *p*")

JUSTIFIED TRUE BELIEF NORM

- Compliance with JTB norm requires

  - Stating only justified & true claims, *or*

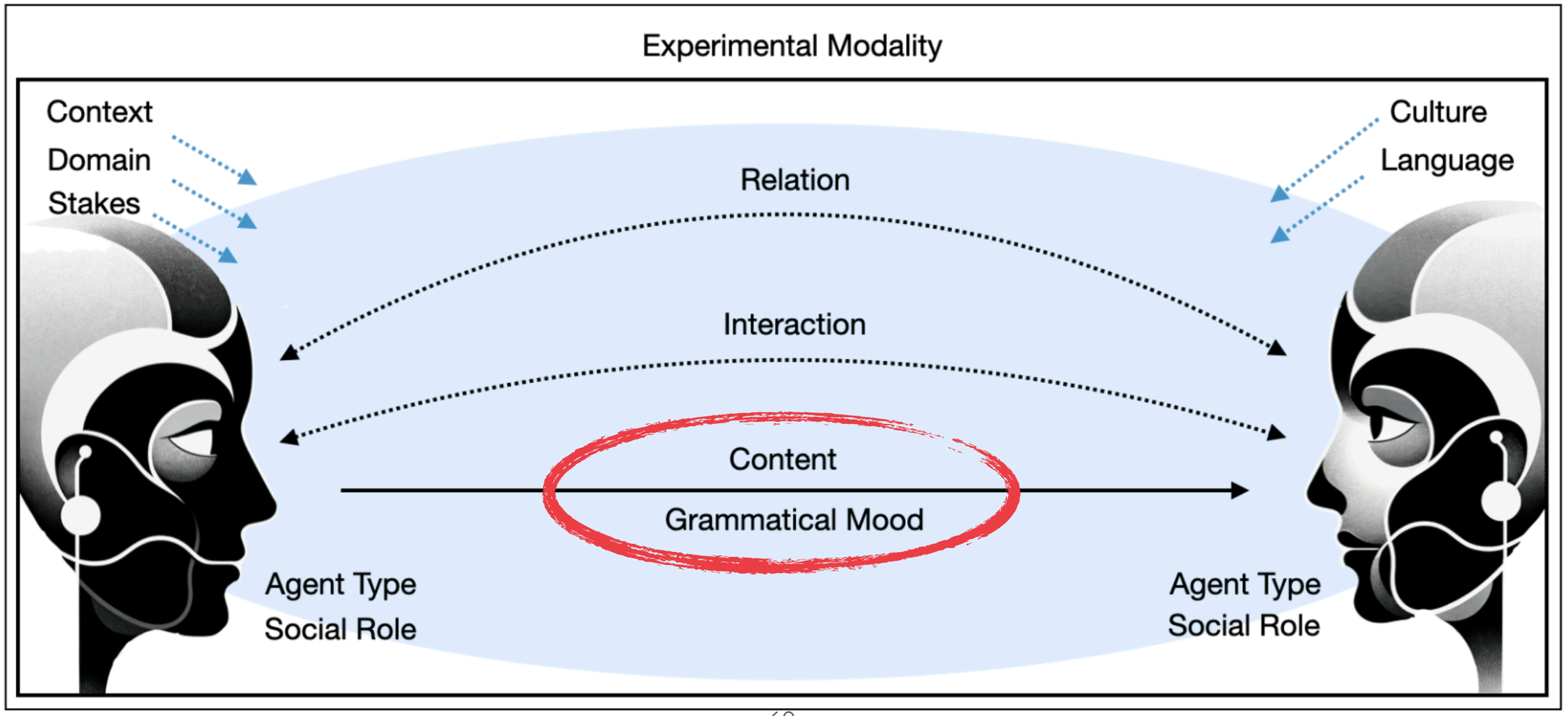
  - Hedging ("might *p*", "probably *p*")

Hard to know the facts.

Easier to know when one doesn't know the facts.

Instead of improving "world-knowledge", we might teach AI to appropriately state uncertainty (cf. Yona et al. 2026).

FACTORS


Experimental Modality
Context
Domain
Stakes
Culture
Language
Relation
Interaction
Content
Grammatical Mood
Agent Type
Social Role
Agent Type
Social Role

EPISTEMIC MODALITY

- Compare:
  - "The flight to Paris departs at Gate 42"
  - "The flight to Paris must depart at Gate 42"
  - "The flight to Paris should depart at Gate 42"
  - "The flight to Paris probably departs at Gate 42"
  - "The flight to Paris might depart at Gate 42"

421

50.1%

∅ 46 (14)

Experimental Design: 2 × 2 × 5 → **20 Conditions**

Agent Type

(2 levels)

- **Human**
- **AI**

×

Epistemic Situation

(2 levels)

- **JFB**
  (Justified False Belief)
- **UTB**
  (Unjustified True Belief)

×

Content / Modality

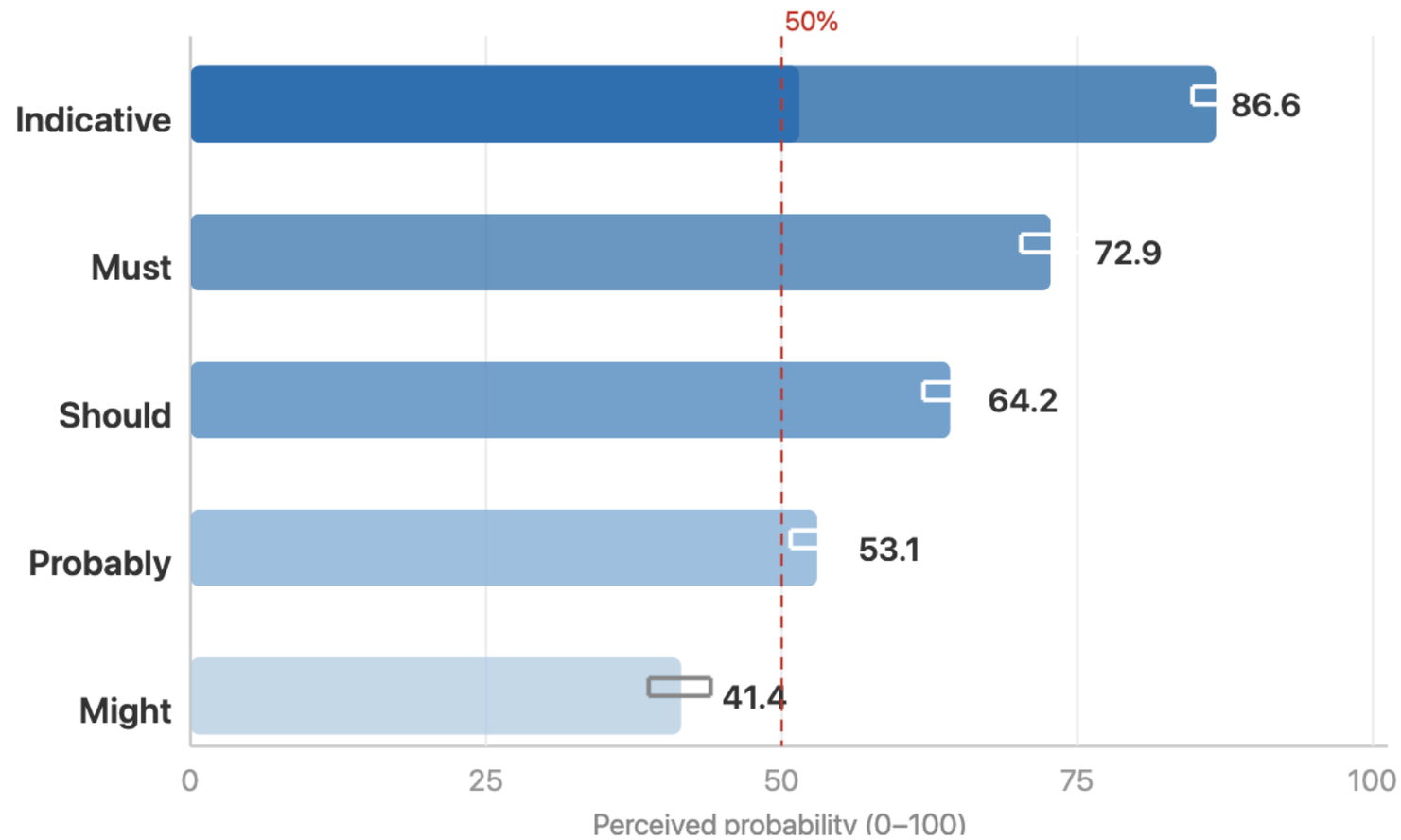
(5 levels)

- **Indicative**
- **Must**
- **Should**
- **Probably**
- **Might**

PERCEIVED PROBABILITY

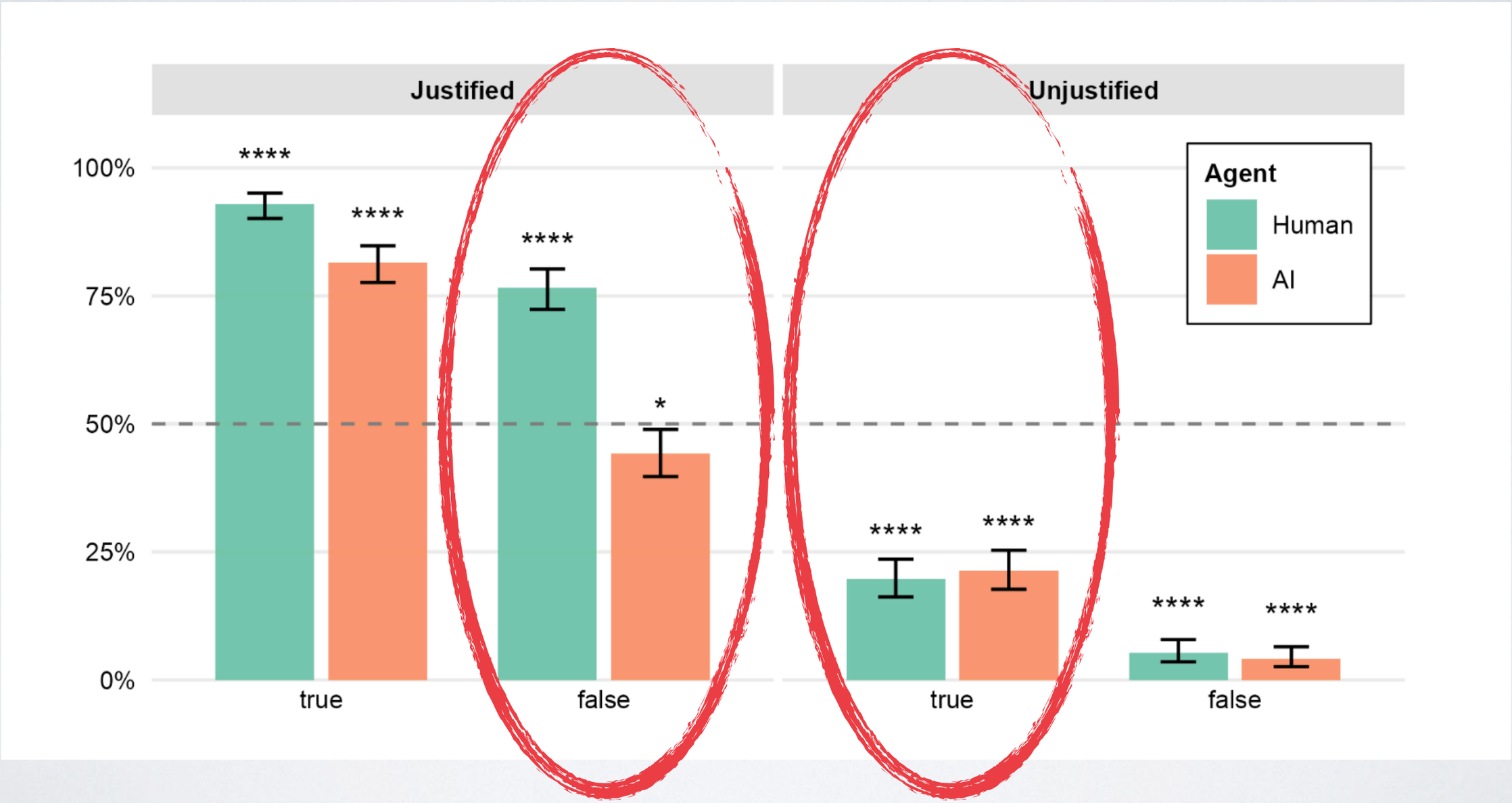
Figure 3 — Perceived Probability by Modal Expression

Mean perceived probability (0–100 slider) ± 95% CI. Within-subjects, N = 421. Ordered by descending mean.

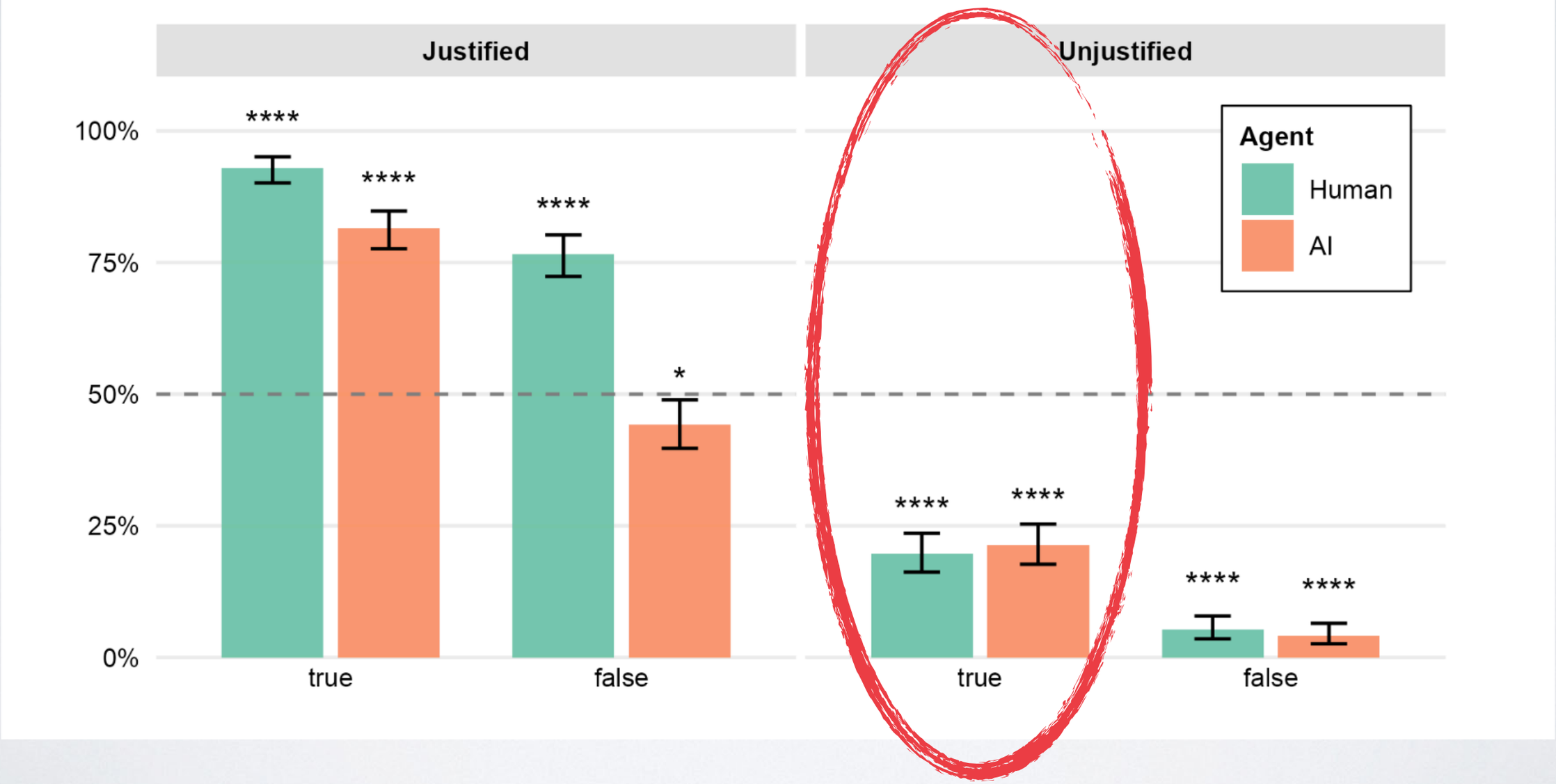


Degen, J., et al. (2019). Definitely, maybe: A new experimental paradigm for investigating the pragmatics of evidential devices across languages. *Journal of Pragmatics, 140*, 33–48.

INDICATIVE

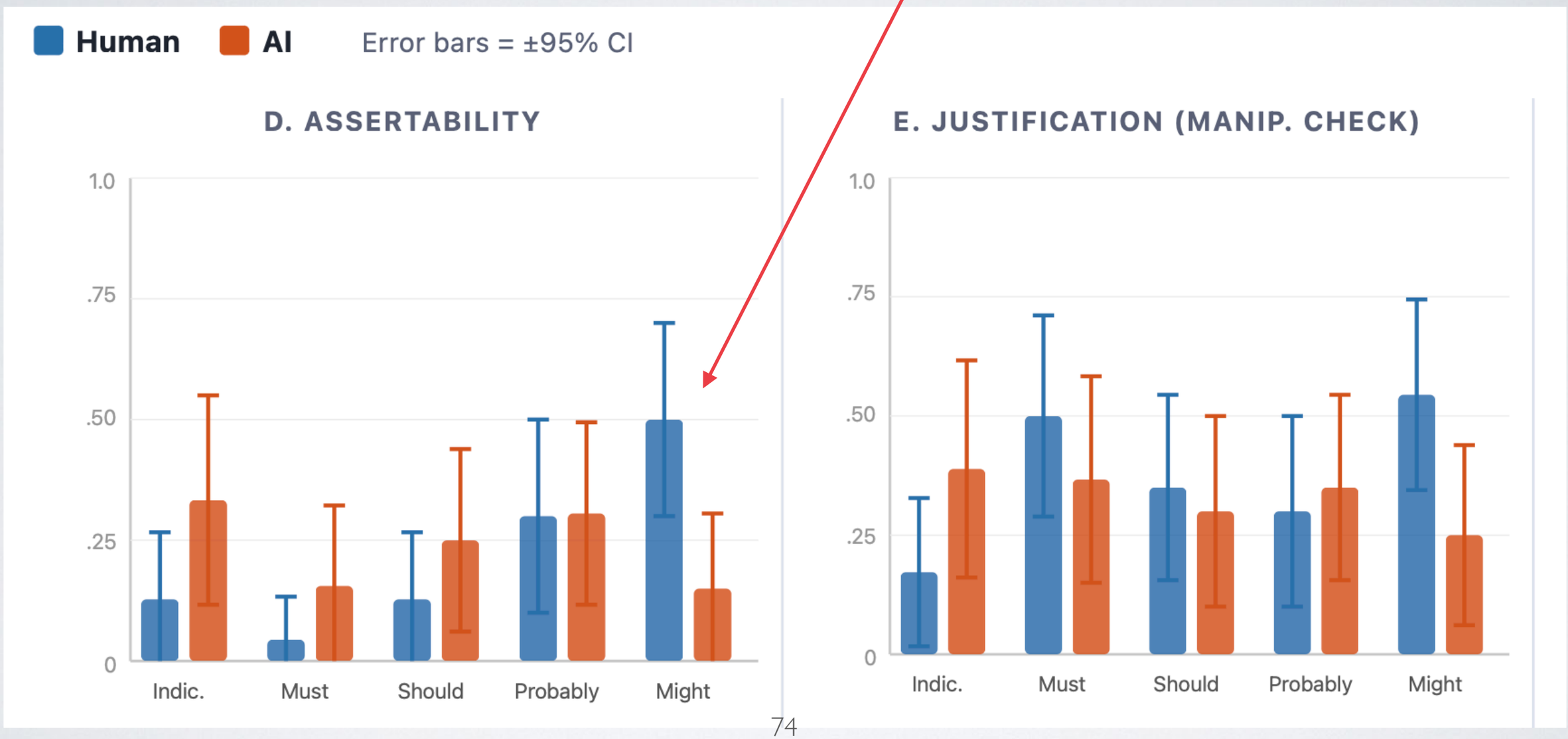

Justified
Unjustified
100%
75%
50%
25%
0%
****
****
****
*
****
****
****
****
true
false
true
false
Agent
Human
AI

INDICATIVE


Justified
Unjustified
100%
75%
50%
25%
0%
****
****
****
*
****
****
****
****
true
false
true
false
Agent
Human
AI

UNJUSTIFIED TRUE BELIEF

If unjustified, even modal claims do not reach assertability beyond chance.

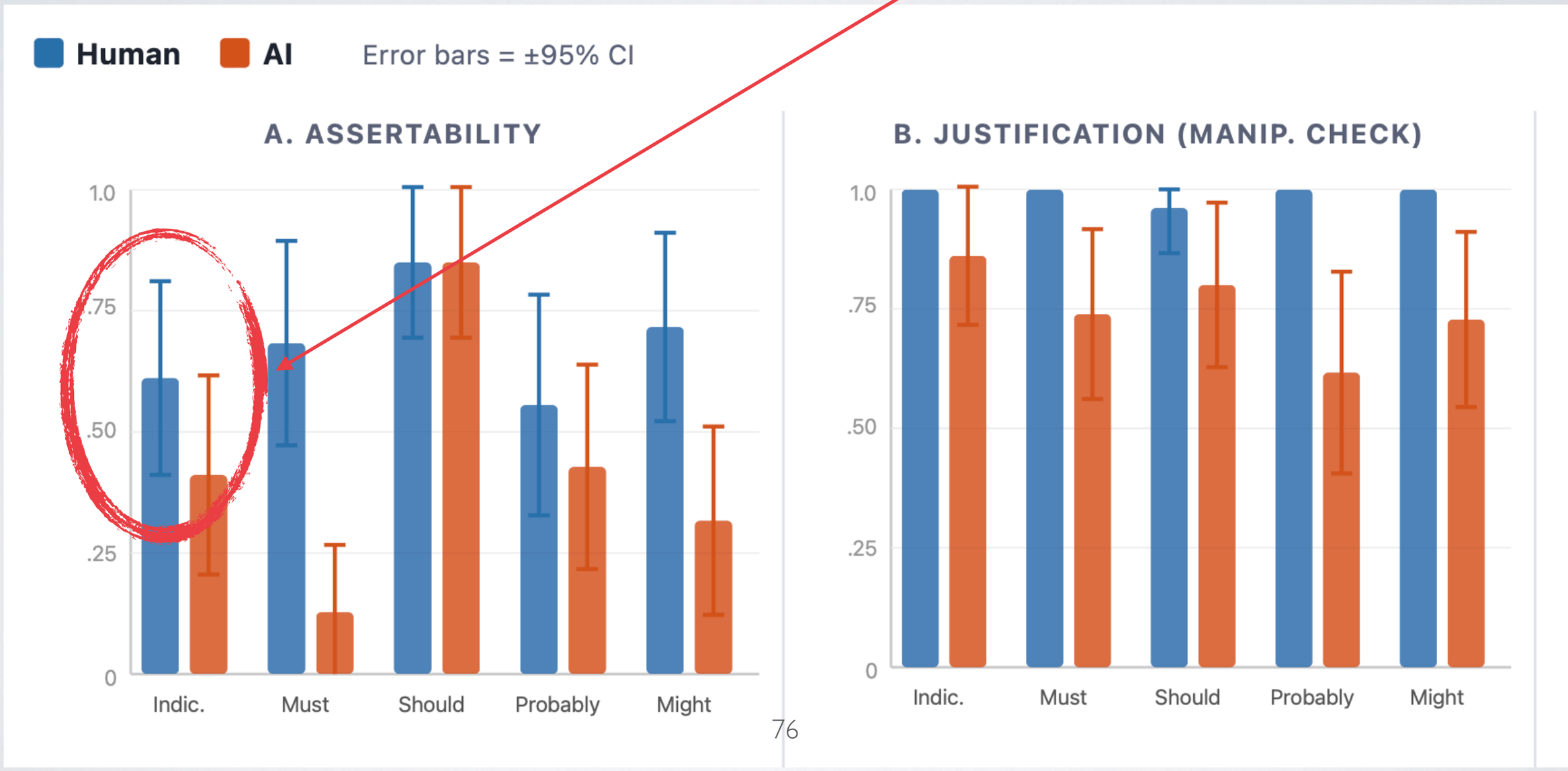


INDICATIVE


Justified
Unjustified
100%
75%
50%
25%
0%
****
****
****
*
****
****
****
****
true
false
true
false
Agent
Human
AI

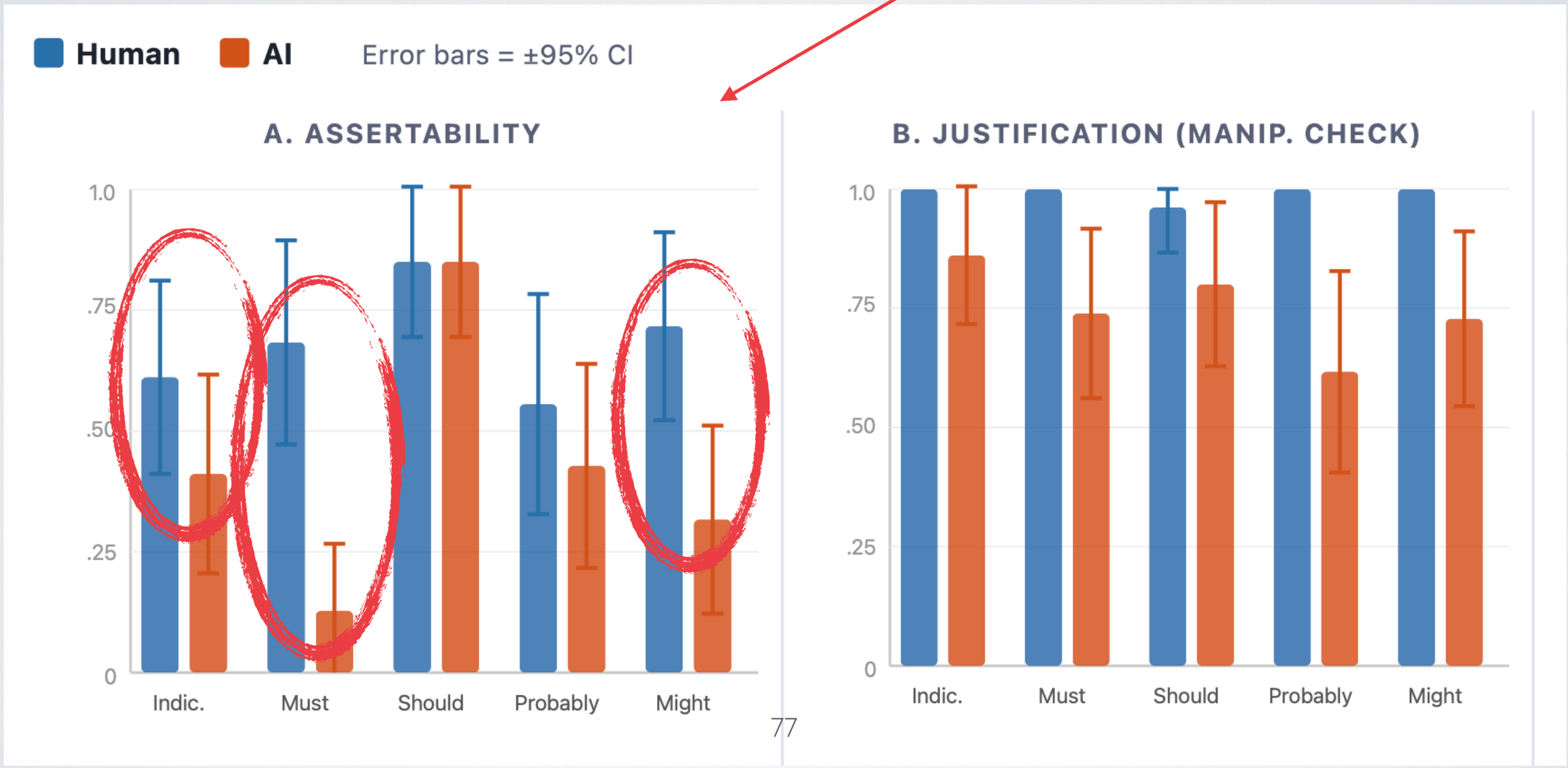
JUSTIFIED FALSE BELIEF

Difference across agents for the indicative replicates



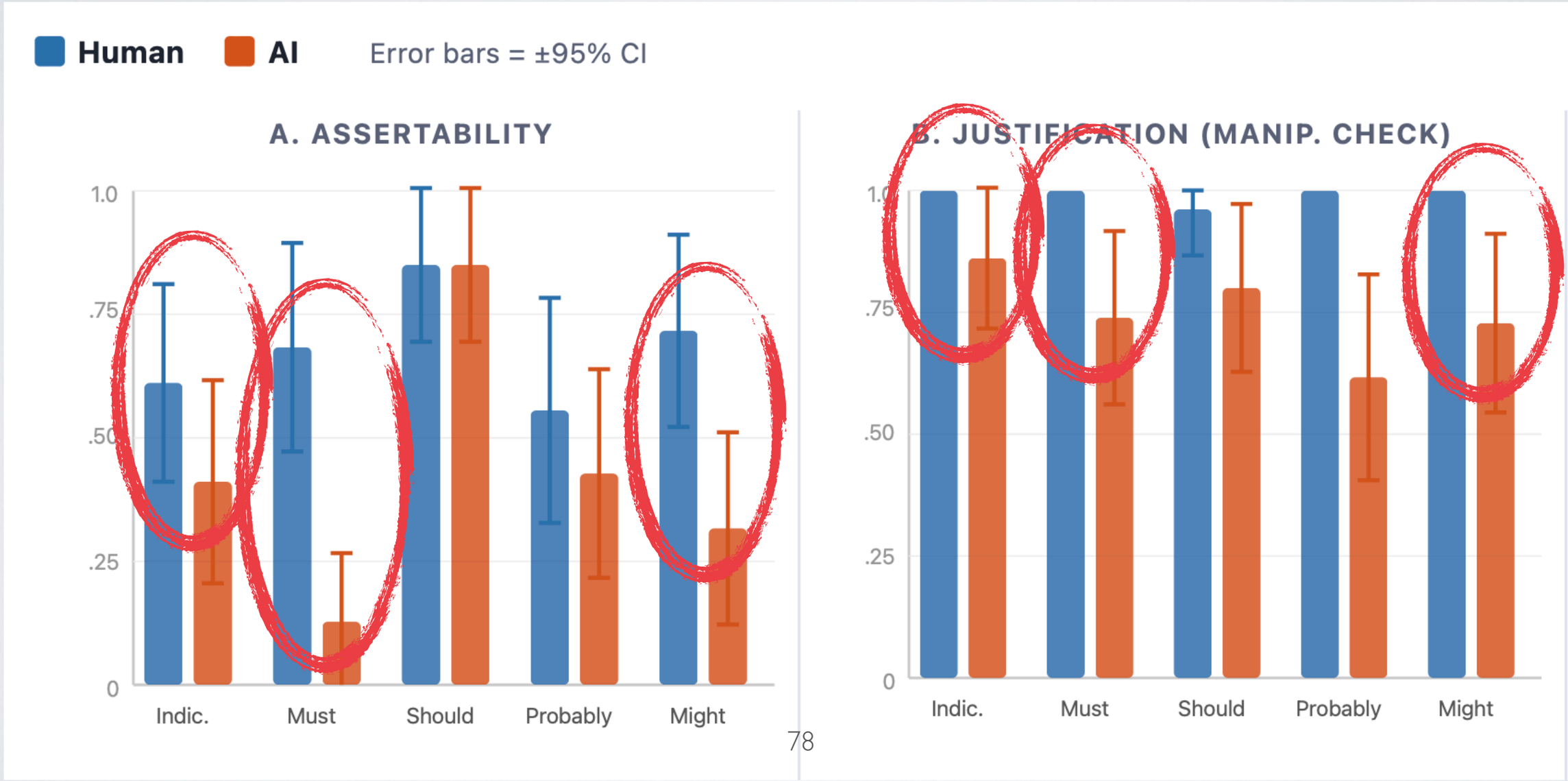
JUSTIFIED FALSE BELIEF

Difference across agents replicates generally



JUSTIFIED FALSE BELIEF

Difference across agents replicates generally
Though partly driven by justification?



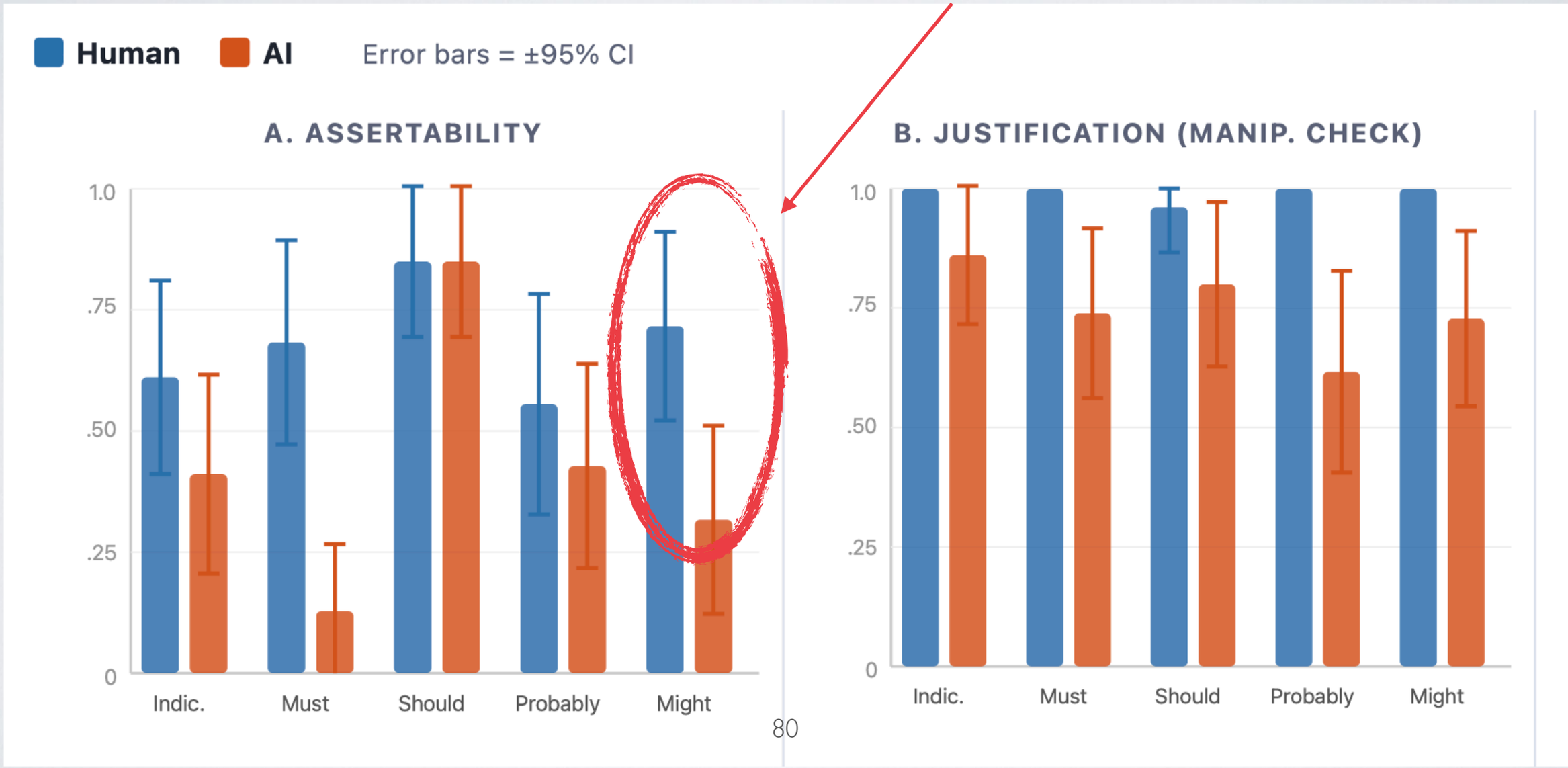
JUSTIFIED FALSE BELIEF

"Should" claims *more* assertable than indicative;
No difference across human v. AI (possibly ceiling effect).



JUSTIFIED FALSE BELIEF

"Might" claims *not more* assertable than indicative;
Big difference across human v. AI



EPISTEMIC MODALITY

- Completely understudied. *The* easiest way to align LLMs (Yona, 2026)
- Highly complex & expression-sensitive:
  - "Should" claims *increase* assertability; *no* difference across H v. AI
  - "Might" claims do *not increase* assertability; big difference across H v. AI
- Assertability of modal claims is selectively agent-sensitive. Key for aligning LLMs.
- Four large-scale studies (N=3200) underway.

SUMMARY

- Humans and chatbots might be subject to different norms of assertion.
  - Whether different norms apply might depend on context (cf. game), though the game study needs refinement.
- Theoretically, the easiest way to align chatbots to what we normatively expect of them is by teaching them appropriate hedging.
  - However, normative intuitions concerning hedged claims are more complex than one might assume & need further work.

THANK YOU!

Comments, criticism, collaboration:
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www.talkingtobots.net